

Composite Si_3N_4 Foils for Suppression of ^{10}B in ^{10}Be Accelerator Mass Spectrometry



M.Sc. Thesis in Physics

Benedikt. P. Kristinsson

202006628

Supervisor: Jesper Olsen

UNIVERSITY OF AARHUS

DEPARTMENT OF PHYSICS AND ASTRONOMY

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Abstract

The effective suppression of the isobar ^{10}B is one of the main technical challenges in ^{10}Be detection via AMS, due to the ions' identical mass-to-charge ratio. Standard separation techniques, such as degrader foils and gas ionisation chambers, involve trade-offs between transmission efficiency and background rejection. At VERA, the Si_3N_4 foil stack method demonstrated effective suppression of ^{10}B at 3 MV, tripling the isotopic-ratio efficiency. In this thesis, we implemented the foil stack method at 1 MV to assess its suitability at AARAMS. Isobar suppression was analysed using two approaches: An initial method based on TRIM/SRIM simulations of ion stopping ranges, and a proposed analytical model incorporating energy loss constraints to define optimal pressure regimes. These analyses yielded pressure values of 1.45 ± 0.25 mbar and 12.2 ± 5.45 mbar for their respective foil thicknesses. Furthermore, an algorithm was developed to optimise foil thickness and detector gas pressure, while a statistical clustering algorithm (OKOH) was introduced to identify the ^{10}Be signal and remove outliers. Due to uncertainty in the actual foil thickness, the experiments at AARAMS were run at lower energies than those simulated. The choice of running at a lower level potentially contributed to the poor data quality. Trials at AARAMS revealed discrepancies between expected and observed performance, indicating that the initial TRIM-based method was not valid at the low energies used. Additionally, there was uncertainty regarding the pressure readings on the GIC, which further complicated the analysis. Based on the measured data, there was insufficient evidence to confirm the method's effectiveness under the tested conditions.

Acknowledgments

I would like to extend my deepest gratitude to my supervisor, Jesper Olsen, for being an excellent mentor and for allowing me to work under his guidance and gain hands-on experience with the AMS. Thank you for your supervision, professional sparring, and for the invaluable experience I gained as part of the AARAMS group.

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Preface

This is my Master's thesis submitted to the Department of Physics and Astronomy at Aarhus University. This work was performed during the spring semester of 2024 until the end of the autumn semester of 2025 under the supervision of Jesper Olsen at the Department of Physics and Astronomy, Aarhus University, 8000 Aarhus C, Denmark.

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Chapter 1

Introduction

For centuries, scientists have sought to determine the age of ancient objects, geological formations, and even the Earth itself. One of the most well-known methods developed for this purpose is radiocarbon dating, which leverages the decay of ^{14}C to estimate the age of organic material. This technique relies on the predictable decay of carbon isotopes over time, which can be modelled using differential equations. However, its effectiveness diminishes for samples older than approximately 50,000 years [1].

To overcome the limitations of ^{14}C , the cosmogenic isotope ^{10}Be , which possesses a much longer half-life, is used and is better suited for dating significantly older materials.

^{10}Be has become invaluable in geology, environmental science, and archaeology. It is produced when cosmic rays interact with the Earth's atmosphere. It is beneficial for dating sediments and other geological records, providing insights into long-term climatic and environmental changes. A key advantage of ^{10}Be is its long half-life of approximately $t_{1/2} = 1.387 \pm 0.0012$ Ma [2, 3], which allows researchers to study timescales far beyond the reach of radiocarbon dating. However, this property presents a detection challenge: Traditional decay counting methods are impractical due to the extremely low decay rate. This difficulty is compounded by the nature of ^{10}Be 's gamma-less beta decay, which has a low Q value of only 556 keV[4], corresponding to the maximum possible beta particle energy. The resulting low-energy beta

particles are challenging to detect, particularly in natural samples with low ^{10}Be concentrations, necessitating the use of more advanced techniques.

Accelerator Mass Spectrometry (AMS) has emerged as a powerful alternative. Rather than detecting decay, AMS directly counts ^{10}Be atoms, enabling high-precision measurements even in microscopic or low-concentration samples. However, AMS detection of ^{10}Be presents its challenge: Interference from the isobaric ion ^{10}B .

^{10}B and ^{10}Be share the same mass number but differ in nuclear properties such as isospin. Their near-identical mass complicates physical separation, making it challenging to distinguish between them. Effective ^{10}B suppression is significant due to its significantly higher natural abundance, which can lead to detector over-saturation and inaccurate results. To minimise contamination, the preparation of ^{10}Be samples is often carried out in a cleanroom environment to avoid the presence of ^{10}B .

While degrader foils and gas-filled detectors have been employed to separate these isobars, they often involve trade-offs between detection efficiency and background suppression. Recent innovations offer promising alternatives. Steier et al. (2019) [5] introduced a technique using a composite silicon nitride (Si_3N_4) foil stack to separate ^{10}Be from ^{10}B at higher acceleration potentials (3 MV).

This thesis investigates whether a similar approach can be applied at lower voltage settings (1 MV), making it suitable for smaller AMS facilities such as AARAMS. This thesis initially set out to validate the proof of concept from an earlier study [5], but several challenges arose during the experimental phase. These included technical issues with the accelerator and its electronics, as well as a design limitation in the sample holder, which affected the fit of the foils. These factors influenced the data collection within the scheduled measurement window, resulting in a more limited dataset than initially anticipated. Consequently, the research approach was adapted accordingly. Additionally, some early hypotheses redirected the focus of the work, ultimately shaping the final scope of the thesis.

1.1 Thesis Structure

The necessary theoretical framework will be introduced for the understanding of the setup for suppressing ^{10}B in ^{10}Be AMS measurements, and the energy loss mechanisms with a focus on the detector.

The experiment's methodology will then be described in detail, covering the AARAMS setup, the two isobar suppression methods (degrader- and stacked foil techniques), and the procedures used for data collection. This chapter also presents the framework for the SRIM/TRIM (Stopping and Range/Transport of Ions in Matter) simulations. Additionally, the focus will be on the algorithm developed for data acquisition, isotope counting, and statistical analysis of the ^{10}Be signal.

The analysis and discussion will begin with the results from the SRIM/TRIM simulations, focusing on the calculated Si_3N_4 thicknesses and pressure for both the initial and proposed methods. It will then proceed to compare the algorithm developed in this thesis with existing approaches for ^{10}Be counting. Finally, the discussion will conclude with the experimental results, evaluating the efficiency of the two experimental setups and the effectiveness of ^{10}B suppression using Si_3N_4 stack foils.

Finally, the results will be summarised, and their broader implications for ^{10}Be AMS measurements will be discussed.

Chapter 2

Theory

This chapter introduces the necessary theoretical background needed to understand the work presented in this thesis.

2.1 Principles of AMS

This section will focus on ^{10}Be and is based on Refs.[6, 7]. The general equipment layout for AMS and the process is illustrated in fig. 2.1, and this section will explain the theoretical knowledge necessary to understand the fundamental processes in AMS.

To understand how AMS can separate isotopes with high precision, we first examine the system's core components.

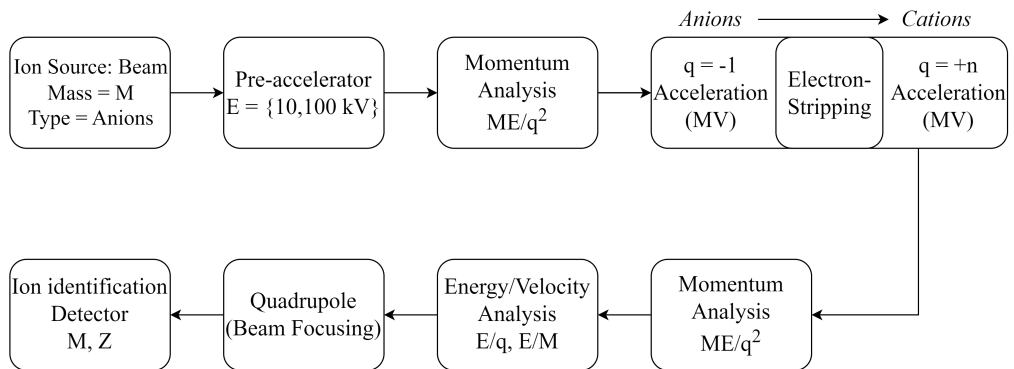


Figure 2.1: Schematic layout of equipment for AMS based on Ref. [7].

Ion Sputter Source

An AMS sputter source typically consists of a system with a heated caesium, Cs, reservoir, where Cs vapour is maintained at a pressure of approximately 10^{-4} mbar. This vapour is pumped into a cavity (see fig. 2.2). The cavity, which includes the ioniser, is constructed from a metal with a low work function, such as tantalum, tungsten, or molybdenum. The low work function refers to the energy required to remove an electron from the metal's interior to a point just outside its surface. This area is heated to 1000°C to thermally ionise the Cs into Cs^+ . The symmetry of the cavity is designed so that Cs^+ ions are focused to strike the sample target. The Cs^+ ions are accelerated by an electric potential applied between the ioniser and the sample target. When these energetic ions strike the target, they cause atoms to be ejected from its surface.

The total energy of an emitted ion depends on the applied voltage, specifically the extraction voltage, U_{ex} , which is set between the ioniser and the sample target. The target voltage, U_{tar} , is utilised to accelerate the ions away from the sample.

The polarity of the applied voltage determines whether we extract anions (negative ions) with a negative voltage or cations (positive ions) with a positive voltage, subsequently accelerating them away from the ion source. The total energy of an emitted ion is given by:

$$E_{\text{source}} = eq(U_{\text{ex}} + U_{\text{tar}}) \quad (2.1)$$

where e is the elementary charge and q is the charged state number [6].

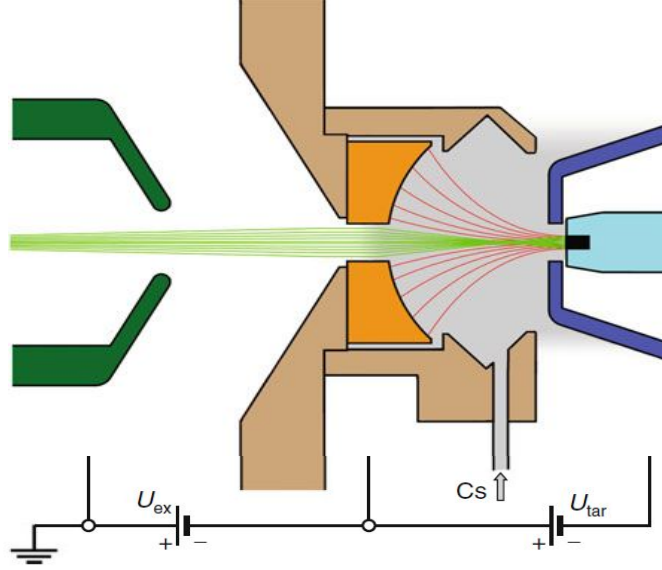
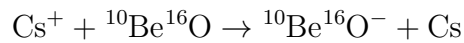


Figure 2.2: A generalised schematic of an ion sputter source for generating negative ions from Ref. [6]. Cs vapour is produced and introduced into the ion source via a tube (grey arrow). On the heated surfaces of the ioniser (orange), Cs atoms are ionised (red lines), which are focused onto the sample (black) within the metal target (cyan) by the target voltage (U_{tar}). As the Cs^+ ions sputter the target, negative carbon ions, Cs^- , (light green) are generated and then accelerated by U_{tar} , followed by the extraction voltage (U_{ex}).

Formation of ^{10}Be Negative Ions

During the ion sputtering process, Cs^+ ions bombard a beryllium oxide ($^{10}\text{Be}^{16}\text{O}$) target, imparting energy to the target molecule. As a result, atoms and molecules from the target surface are ejected. In addition to the ejected neutral atoms, a variety of secondary ions are generated. Some of the Cs^+ ions will induce electron transfer from the target, leading to the formation of negatively charged ions such as $^{10}\text{Be}^{16}\text{O}^-$. The process of electron transfer will look like the following:



This electron transfer process is a key mechanism for generating the ^{10}Be ion beam used in AMS measurements.

Magnets

The beamline of charged Ions, $X^q = {}^{10}\text{Be}^{16}\text{O}^-$, has a charge of $q = -1$, and therefore, a dipolar magnetic field is needed if they are to be moved along the AMS setup. Commonly known as a bending/injection (BI) magnet used for moment analysis, it selects the mass to be analysed as it enters or is injected into the accelerator.

When a charged particle moves perpendicular to a magnetic field, it experiences a magnetic force that causes it to travel in a circular trajectory known as the Lorentz force, $\vec{F} = q(\vec{v} \times \vec{B})$. The radius of the circular trajectory is determined by equating the centripetal force with the Lorentz force and utilising the kinetic energy and velocity of the charged particle, $E_{\text{kin}} = E_{\text{source}} = \frac{1}{2}Mv^2$ and $v = \sqrt{\frac{2E_{\text{kin}}}{M}}$ respectively (classically). When analysing the circular motion of ${}^{10}\text{Be}^{16}\text{O}^-$ ions in a magnetic field, relativistic effects would primarily modify the equations by transforming the ion's rest mass, m_0 , to its relativistic mass, m' :

$$m' = \gamma m_0; \quad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where γ is the Lorentz factor. For our experimental setup, this correction is negligible because:

- The ion velocities ($v \ll c$) result in $\gamma \approx 1$
- The kinetic energies ($E_{\text{kin}} \approx 35 \text{ keV}$) are well below the threshold for significant relativistic effects

Therefore, for an charged ion, $X(q, p)$, with charge q and momentum $p = \sqrt{2AE}$ in a homogenous magnetic field, B , the radius, r , of the curvature of its trajectory can be determined by analysing these relationships yielding

$$\left(\frac{M}{q}\right) \left(\frac{E_{\text{kin}}}{q}\right) = k_1 (Br)^2 \quad (2.2)$$

where k_1 is a proportionality constant. An energy/velocity analyser, often placed after the acceleration section of the beamline, is also known as an Electrostatic Analyser (ESA). The ESA is an instrument that utilises electric fields to separate charged particles based on their energy per charge. The force on a charged particle in an electrostatic field can be applied with the work-energy theorem to determine the trajectory of the curvature imposed by the electrostatic field.

$$\frac{E}{q} = k_2 \mathcal{E} r. \quad (2.3)$$

where k_2 is a proportionality constant.

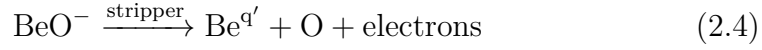
As the charged ion beam travels through the system, even minor angular deviations and energy differences among the ions cause the beam to diverge due to the Lorentz force. Quadrupole magnets are used to correct this and maintain beam focus. These magnets produce a field gradient that focuses the beam in one transverse direction while defocusing it perpendicularly. By configuring several quadrupoles in a particular order, it is possible to achieve focusing in both planes, effectively collimating the beam while maintaining transmission efficiency.

Tandem-Accelerator

A tandem accelerator utilises a two-stage acceleration process, which involves converting anions into cations. The device accelerates ions through an electric potential difference, which is achieved by creating a high-voltage terminal at the midpoint of the accelerator. This allows for the efficient acceleration of particles to higher energies.

In the first stage, anions are injected into the accelerator with an initial energy $E_i = V_i e$, where V_i is the injection voltage and e is the electronic charge. As the anions are accelerated towards the high-voltage terminal, they pass through an electron stripper - usually a gas cell filled with argon (Ar) set at 10^{-2} mbar. The electron stripper removes electrons from anions, converting them into cations through scattering reactions with Ar.

This process increases the charge, $q \rightarrow q + n = q'$, resulting in a higher charge state. Additionally, the stripper dissociates molecular ions, which helps eliminate molecular isobars, as in the reaction:



After passing through the electron stripper, the cations are further accelerated by the terminal's high-voltage potential V_t . This results in an increase in the kinetic energy of the newly acquired cations.

The total energy E_f of the cations after acceleration is given by:

$$E_f = E_i + (q' + 1)eV_t \quad (2.5)$$

This equation shows that the final energy depends on the ions' initial energy, E_i , and the terminal voltage V_t .

For a more generalised expression of the cation energy E_f after acceleration in the tandem accelerator, we can use:

$$E_f = \left(\frac{(V_i + V_t)M_p}{M_i} + q'V_t \right) e \quad (2.6)$$

Where $V_i e$ is the initial energy of the injected anions, V_t is the terminal voltage, e is the electronic charge, q' is the charge state of the cations after stripping, M_p is the mass of the cation $^{10}\text{Be}^+$, and M_i is the mass of the anion $^{10}\text{Be}^{16}\text{O}^-$.

2.2 Energy Loss Mechanisms in a Gas Ionisation Chamber

The detection mechanism in a Gas Ionisation Chamber (GIC) is based on the principle of energy loss of charged particles as they traverse through a gas. The stopping power quantifies the energy loss of an ion, $\frac{dE}{dx}$, which represents the rate at which energy is transferred from the ion to the gas medium per unit length.

Most of the ion's energy is used to produce electron-ion pairs in the gas, a process known as electronic energy loss, while only a negligible fraction is converted to thermal energy. This energy loss mechanism is described by the Bethe-Bloch equation [8], which quantifies the stopping power of a charged particle as it travels through matter. The underlying interaction arises from the electromagnetic field of the moving ion interacting with the electrons in the medium. The electric field magnitude E_{field} generated by a particle with charge Ze at an impact parameter b is given by:

$$E_{\text{field}} = \frac{Ze}{4\pi\epsilon_0} \frac{1}{b^2} \quad (2.7)$$

By integrating over all relevant impact parameters, the total energy loss per unit length is calculated by summing the energy transferred to electrons at various distances. In GICs, isobutane (C_4H_{10}) is often chosen as the working gas because its low ionisation energy and high ionisation cross-section make it highly effective for producing detectable signals. For C_4H_{10} , where n is the electron density, the number of electrons within a radius b and thickness db is $2\pi b db n$. This leads to the following non-relativistic expression for the energy loss:

$$\left(\frac{dE}{dx}\right)_e = \frac{4\pi Z^2 e^4}{m_e v^2} n \ln \left(\frac{2m_e v^2}{I}\right) \quad (2.8)$$

In this equation, v is the velocity of the ion, m_e is the electron mass, and I is the mean excitation energy of the gas. The energy loss as described

by the Bethe-Bloch equation varies with the ion's velocity, scaling as $\frac{Z^2}{E}$ at high energies and decreasing at low energies due to incomplete ionisation of the ion.

High and Low Energy Regimes

In the relativistic regime (high energies), the energy loss becomes inversely proportional to the ion energy, scaling as $\frac{Z^2}{E}$, where E is the energy of the ion. At these speeds, energy loss primarily depends on the ion's velocity, i.e. $\frac{dE}{dx} \approx f(v)$, with charge having a less significant effect.

The situation is more complex at lower energies, as ions may not be entirely stripped of their electrons. This incomplete stripping results in a fluctuating charge state, reducing energy loss compared to the high-energy regime. In this regime, the energy loss depends on several factors, including the ion's velocity v , charge state q , the electron density ρ of the medium, and the interaction cross-section σ , such that $\frac{dE}{dx} \approx f(q, v, \rho, \sigma)$.

Ion Identification and Detector Calibration

The energy loss mechanisms described by the Bethe-Bloch equation are crucial for identifying ions based on their nuclear charge, Z . As ions traverse matter, their energy loss depends strongly on Z^2 , making this an effective method for distinguishing ions with similar mass but different charge states. By measuring their energy loss profiles, GIC detector can differentiate ions such as $^{10}\text{Be}(Z = 4, N = 6)$ and $^{10}\text{B}(Z = 5, N = 5)$, which have identical mass numbers but distinct energy loss behaviours due to their differing nuclear charges.

Chapter 3

Methodology

3.1 AMS at Aarhus University

This section describes the setup of AARAMS, covering the ion source, beamline, acceleration process, and detection system.

Conventional mass spectrometry is sufficiently sensitive to provide adequate measurements of isotopic ratios of stable nuclei, such as isotopes [7, 9]. However, to measure ^{10}Be , with isotopic ratios as small as 10^{-16} , the limitations of conventional mass spectrometers are exceeded. Utilising AMS allows access to higher energies, which helps resolve ambiguities in identifying nuclei, atoms, and molecular ions of the same mass. As described in section 2.1, the setup at Aarhus University follows a similar configuration. The AMS system for AARAMS comprises several components, which will be described in this section and their effects on the ion beam. The schematic layout of AARAMS, depicted in fig. 3.1, shows that it can be divided into five main sections: Ion source, Low Energy section (LE), 1 MV accelerator, High Energy section (HE), and the Rare Isotope section (RI) which includes the detector.

Additionally, the optical beam-handling components include Einzellenses, Y- and X-steerers, X-slits, apertures, Q-snout lenses, and Q-pole lenses, as

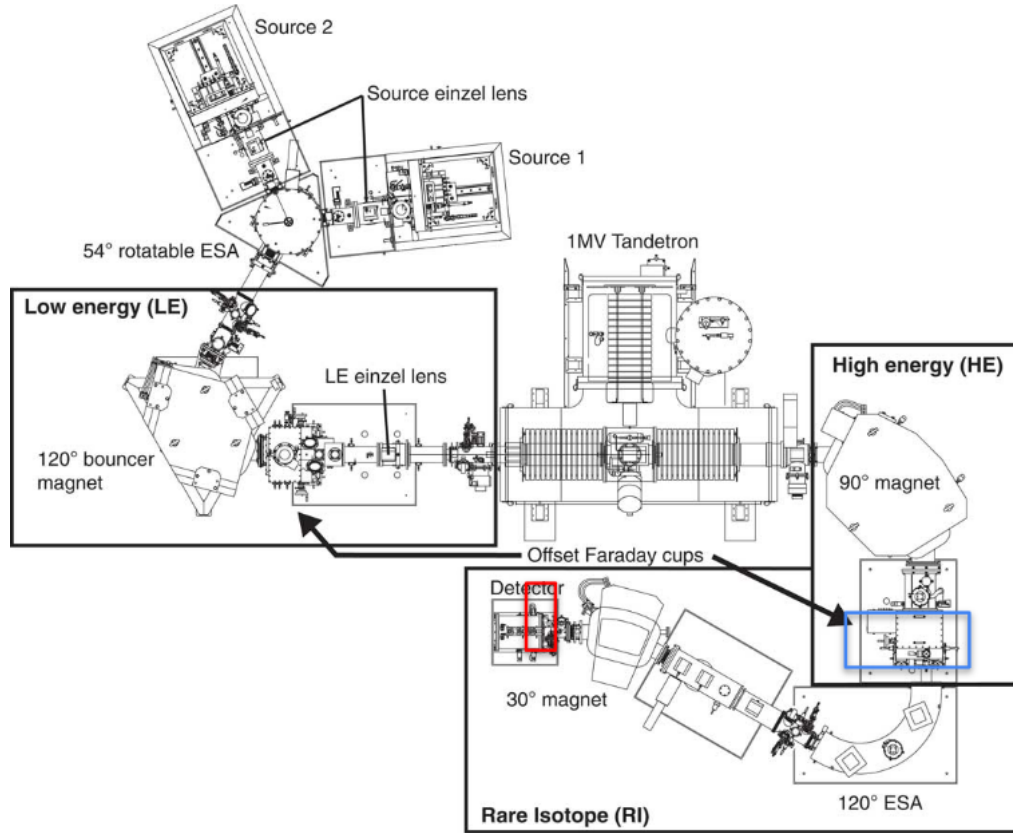


Figure 3.1: A 5-sub division of the 1 MV Tandetron Accelerator layout at Aarhus University Ref.[10]. The blue rectangle is where foils can be slit down to intersect the beam. The red rectangle is where we will add Si_3N_4 foils, as suggested by Ref. [5]. The latter rectangles represent two different setups.

well as retractable Faraday cups (FCs) for measuring and analysing the beam current.

Ion Source

The first section in the AARAMS is where the beam of $^{10}\text{Be}^{16}\text{O}^-$ and other anions is created. The AARAMS uses a caesium sputter source, as described in section 2.1. It has two independently operable negative ion sources, HVEE SO-110C-1 (source 1) and SO-110C-2 (source 2), each capable of holding up to 50 samples. This allows for a quick switch between samples, which is practical.

The AARAMS ion source (section 2.1) operates at $U_{\text{ex}} + U_{\text{tar}} = 35 \text{ kV}$, where U_{tar} is set at 7 kV and U_{ex} set at 28 kV, with Cs vapour heated to 115 °C and an ionisation cavity at 1000 °C, which is achieved by passing an current of 18 A through its embedded coil. The adjustable sample holder optimises sputtering efficiency.

The beam of anions, consisting of $^{10}\text{Be}^{16}\text{O}^-$, is subjected to uniform acceleration, resulting in equal kinetic energy for each anion. This ensures that all anions in the beam, regardless of their mass or charge, attain the same kinetic energy under these conditions. They are accelerated towards a rotatable ESA.

The ESA is a 54 ° electrostatic analyser with a radius of 469 mm, a gap of 53 mm, and an energy-to-charge ratio of 44 kV [11]. The ESA can be seen in fig. 3.2, where two curved, parallel conducting plates create a radial electric field of strength $\mathcal{E}$ between them.

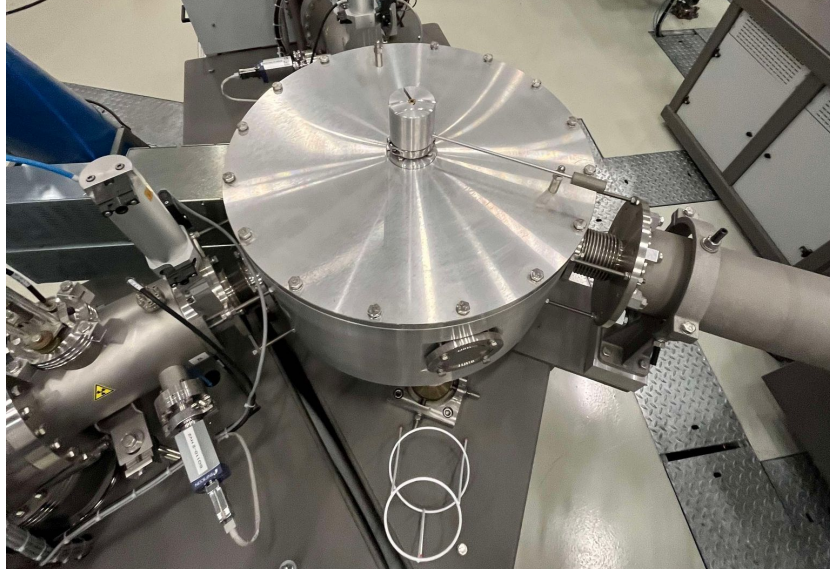


Figure 3.2: Rotatable 54° Electrostatic analyser.

The ESA guides ions along a circular path, allowing only those with specific energy levels to pass by setting the arc radius, r , according to the ions' energy-to-charge ratio, as described in eq. (2.3). The electrodes are held at equal voltages but with opposite polarities [11]. Given the electrostatic

rigidity of an ion, the required voltage on one of the ESA electrodes can be calculated as:

$$\mathcal{V}_{\text{ESA}} = \frac{E_i x_{\text{gap}}}{q r} \quad (3.1)$$

where the ion's energy, E_i , the gap between the electrodes, x_{gap} , the ion's charge, q , and the radius of curvature, r , are used in the calculation.

LE section

Anions are sorted and directed in the LE section based on their energy and charge before entering the accelerator. The crucial elements here include the ESA and the bouncer magnet, which help separate and direct anion beams of different masses towards the accelerator.

To ensure that anion beams of different masses follow the same trajectory through the bouncer magnet, we must adjust the bouncer voltage to provide uniform magnetic rigidity for all anions in the beam. The relationship between two ion beams, one of reference mass and the other of a different mass, depends on the equality of their magnetic rigidity, as seen with ^9Be and ^{10}Be .

$$\frac{p(^9\text{Be})}{q(^9\text{Be})} = \frac{p(^{10}\text{Be})}{q(^{10}\text{Be})}$$

And by substituting the momentum as defined in section 2.1, we can determine how the bouncer voltage should be adjusted to ensure the same path for both beams.

$$\sqrt{\frac{2m_9 E_9}{q(^9\text{Be})^2}} = \sqrt{\frac{2m_{10} E_{10}}{q(^{10}\text{Be})^2}} \quad (3.2)$$

This equation can be used to tune the bouncer voltage settings such that ion beams of different masses, ^9Be and ^{10}Be , will leave the bouncer magnet following the same trajectory.

Hall effect and magnetic monitoring

To ensure precise magnetic control, the system uses Hall probes, which measure the field strength based on the Hall effect, and are implemented at AARAMS. These calibrated sensors utilise the Hall effect to monitor the strength of the magnetic field within the magnet. The Hall effect occurs when a magnetic field is applied perpendicular to a current flowing through a conductor, causing a Lorentz force to deflect the electrons and accumulate charge on one side of the conductor. This accumulation results in a voltage difference, called the Hall voltage, proportional to the magnetic field strength [12, 13].

The Hall voltage V_H is given by

$$V_H = \frac{IB}{ned_c} \quad (3.3)$$

where I is the current through the conductor, B is the magnetic field strength, n is the charge carrier density, e is the electron charge, d_c is the thickness of the conductor.

By continuously measuring the Hall voltage, we can monitor the magnetic field in real-time, ensuring that it remains at the correct strength to direct the ion beam along its intended trajectory. This is particularly important for ensuring that the beam of anions, such as ${}^9\text{Be}^{16}\text{O}^-$ and ${}^{10}\text{Be}^{16}\text{O}^-$, is precisely focused as it exits the bouncer magnet and proceeds to the accelerator.

To match the trajectory of beams with different masses, the bouncer voltage V_{BNC} must be tuned accordingly [11]. The reference beam, with mass m_{REF} , has a reference bouncing voltage V_{REF} , which is typically set to zero. For a beam of mass m_{BNC} , the energy equation can be expressed as:

$$m_{\text{BNC}}q(V_{\text{Tar}} + V_{\text{Ex}} + V_{\text{BNC}}) = m_{\text{REF}}q(V_{\text{Tar}} + V_{\text{Ex}} + V_{\text{REF}}) \quad (3.4)$$

Isolating for V_{BNC} , we get:

$$V_{\text{BNC}} = (V_{\text{Tar}} + V_{\text{Ex}} + V_{\text{REF}})\frac{m_{\text{REF}}}{m_{\text{BNC}}} - (V_{\text{Tar}} + V_{\text{Ex}}) \quad (3.5)$$

1 MV Accelerator Section

Figure 3.3 provides a view of the 1 MV tandem accelerator at AARAMS, highlighting the entry point for the anion beam.

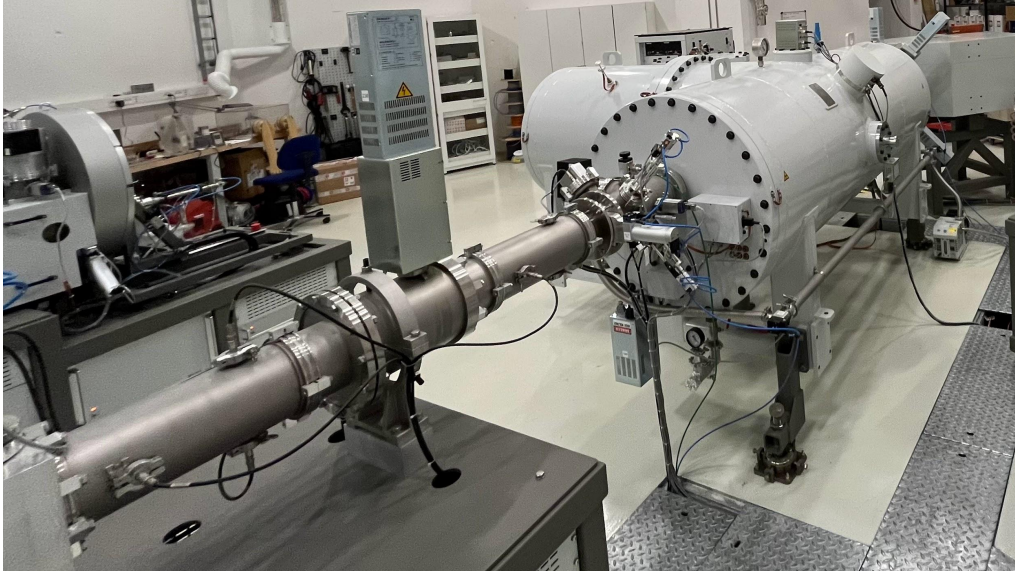


Figure 3.3: The 1 MV tandem accelerator at AARAMS, showing the entry pipe where the ion beam enters the accelerator.

To understand the accelerator at AARAMS, it is essential to be familiar with its main components: An insulating tank filled with SF_6 gas, low- and high-energy accelerator tubes, a Cockcroft-Walton voltage multiplier, and a charge-exchange chamber equipped with gas, pressure gauges, and various beam-handling elements, including Q-snout and Q-pole lenses. At the core of the AARAMS accelerator lies a gas-exchange tube, referred to as the stripper in section 2.1, which enables the dual acceleration of ions within a single voltage potential.

The stripper dissociates $^{10}\text{Be}^{16}\text{O}^-$ and converts the anions into cations by altering their charge state. This process enables the filtering out of smaller molecular fragments in subsequent stages, although it typically reduces the overall beam intensity due to charge-state transformation [14].

At the centre of the accelerator chamber, the terminal acceleration

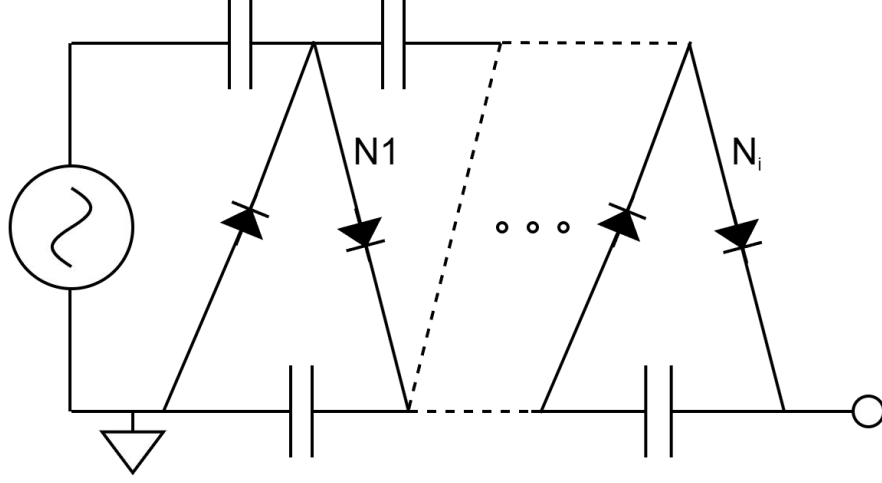


Figure 3.4: Circuit diagram of an N-stage Cockcroft-Walton multiplier, inspired by Refs. [15, 16].

potential is generated by a Cockcroft-Walton (CW) multiplier, which is schematically depicted in fig. 3.4.

The CW multiplier converts a low AC input to a high DC output by employing a sequence of diodes and capacitors, which sequentially step up the voltage. This design ensures a stable and high terminal voltage for ion acceleration. In the CW multiplier, each N stage consists of two capacitors and two diodes. The capacitors are charged in parallel, with every second capacitor connected in series between the ground and the output in each stage. This configuration produces an output voltage proportional to the number of stages, effectively multiplying the input voltage by an even factor.

The output voltage, V_{out} , is therefore given by:

$$V_{\text{out}} = 2N_i V_{\text{in}} \quad (3.6)$$

where N_i is the number of stages and V_{in} is the input voltage.

This terminal voltage creates the electric field that drives ion acceleration throughout the system, with field strength gradually decreasing at the accelerator's extremities.

Charge-exchange gas cell

At AARAMS, the gas used in the stripper is argon, with the option to switch to helium via the dual gas stripper setup [17]. In the stripping process, the anion beam is directed through a gas layer with an areal density of approximately $1 - 2 \mu\text{g}/\text{cm}^2$, while minimising gas diffusion into the low- and high-energy accelerator tubes.

Excessive gas in these tubes can scatter the ion beam, potentially producing neutral particles, cations, and electrons that disrupt transmission efficiency, especially if the vacuum is suboptimal. The low-energy tube is more susceptible to such interference, as the slower ions present are more prone to charge exchanges outside the stripper. To address these issues, turbo-pumps maintain a high vacuum in the tubes, recirculating gas back to the stripping chamber.

If electrons produced through scattering are not directed to a tube electrode, they may gain enough energy to cause further collisions, leading to additional electron or ion production. The resulting X-rays can ionise the insulating gas, drawing charge from the multiplier and terminal rings to the tank wall, leading to voltage drops, known as electron loading. To control radiation levels, a bias ring set to -4 kV is placed at the accelerator entrance to block incoming electrons.

In the LE section of the accelerator tube, ions reach a total energy at the stripper defined by

$$E_{\text{strip}} = E_i + eV_{\text{TV}} \quad (3.7)$$

where E_i is the initial ion energy, and V_{TV} is the accelerator's terminal voltage. Upon entering the stripper, ions undergo two main processes: Electron transfer, which decreases the ion's charge state q by one as an electron is gained, and electron loss, where the ion loses an electron, increasing its charge state. The reactions represent these processes:

$$Z^q + e^- \rightarrow Z^{q-1} \quad \text{or} \quad Z^q \rightarrow Z^{q+1} + e^- \quad (3.8)$$

Each process has a specific probability based on cross-section, ion charge, excitation state, and kinetic energy. As ions move through the gas, they alternate between capture and loss, stabilising into a quasi-pseudo-equilibrium charge state. This process allows various charge states for ^{10}Be , such as $^{10}\text{Be}^{q=+1}$ and $^{10}\text{Be}^{q=+2}$. This schematic is shown in fig. 3.5.

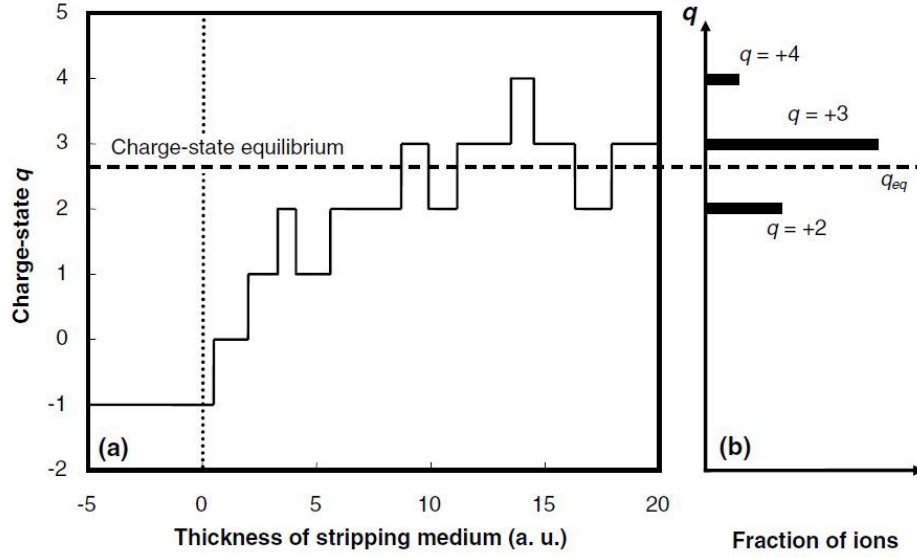


Figure 3.5: Stepwise exchange is shown schematically for an ion of charge $q = 1$ Ref. [18]. The vertical axis represents the charge state, the left horizontal axis shows the thickness of the stripping medium in arbitrary units, and the right horizontal axis indicates the fraction of ions in each charge state.

Although the stripper is a minor component of the AARASM, it plays a crucial role and requires precise handling. Stripper gas pressure is pivotal: It influences the efficiency of molecule ion dissociation, but if too high, it may cause beam divergence due to angular scattering [11]. This effect can be mitigated by the Q-pole lens at the exit.

Optimising the stripper gas cell pressure is essential to suppress unwanted background while maximising the desired charge states of $^{10}\text{Be}^{q=+1}$. However, specifying the exit charge state with complete certainty is challenging due to the energy-dependent charge state distribution, the quasi-pseudo-equilibrium state, and unavoidable scattering events.



Figure 3.6: HE section at AARAMS from High Voltage Engineering Europa B.V.

HE Section

The high-energy cation beam with energy given by eq. (2.6) leaves the accelerator. It now enters the 90° analysing magnet, with a radius of 850 mm, a gap of 50 mm, and a mass-energy product of 63-73 AMU – MeV in the HE section. The 90° HE magnet is shown in fig. 3.6. The increased energy of the cation beam now requires the magnetic and electric fields in filtering components, such as the ESA, to be stronger than those in the preceding sections. When the HE ESA has adjusted the electrostatic rigidity of the incoming ions, the beam is guided through a 120° bouncer magnet. This magnet has a bending radius of 455 mm and a magnetic gap of approximately 62 mm. By controlling the magnetic rigidity, whose ratio is given by eq. (2.2), the trajectory of the charged particles is determined. This magnetic filtering step further separates species based on their momentum-to-charge ratio. The HE magnet analyses the cation beam based on a radius that can be isolated from eq. (2.2). The function of the magnet is to guide the trajectory of $^{10}\text{Be}^q$ and $^{10}\text{B}^q$. The $^9\text{Be}^q$ will be separated and measured

in offset FCs after the magnet. The cation beam will then continue toward the RI section. Additionally, the FC used to measure the ^9Be -cation beam is equipped with two vertical slits (electrodes) that provide information about the beam's position within the cup. The signal generated by these slits can be utilised in a feedback loop to stabilise the terminal voltage of the accelerator, ensuring that the rare isotope beam enters the ESA in the RI section at a precisely defined and reproducible location [11].

Degrader technique

The Degrader technique is a well-established method for isobar suppression of ^{10}B from ^{10}Be . In this approach, a thin silicon foil is placed in the beamline to selectively slow down ions based on their atomic properties and help filter ^{10}B out of the ^{10}Be cation beam. At AARAMS, this technique utilises Si_3N_4 foils, where ions emerge with a distribution of charge states after passing through the foil. Most of these charge states settle at $q = +1$, approximately 55%. The method has been optimised to achieve an isotopic ratio efficiency of around 16%, ensuring reliable isobar separation.

RI Section

The last section before the detector is the rare-isotope section. The main components the ion beam will traverse are the 120° ESA and the 30° magnet before reaching the ionisation chamber section (GIC). We use an ESA with a deflection angle of 120° , a radius of 650 mm, a gap of 25 mm, and an energy-to-charge ratio of 1.5 MV; however, although this ESA removes many unwanted particles, some isobars may still pass through due to increased energies. To further refine the beam, it is directed through a second ESA before passing into a 30° magnet with a radius of 850 mm, a gap of 40 mm, and an energy-to-charge ratio of 63 AMU- MeV, with a maximum tolerance of 73 AMU- MeV. This additional magnetic filtering step ensures that we obtain even greater suppression of background particles and unwanted ions that might have scattered off the ESA electrodes, thereby enhancing the purity of the final beam.

Detector

The GIC detector used in this experiment measures the energy loss of ions as they pass through a chamber filled with isobutane gas, as illustrated in fig. 3.7. When the cation beam enters the detector through a thin foil-covered window, the cations lose energy by ionising the gas and producing electron-ion pairs. An electric field separates these pairs, with electrons drifting toward the anodes and ions toward the cathode. The number of electron-ion pairs created is proportional to the energy lost by the ions.

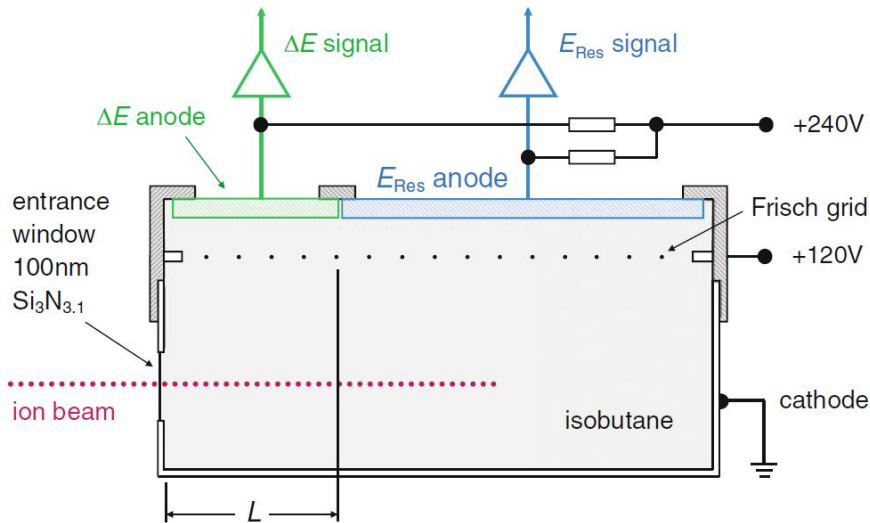


Figure 3.7: Schematic of the ionisation chamber detector, showing the two-part anode and the Frisch Grid positioned between the ionisation region and the anodes. The sensitive volume beneath the ΔE -anode is defined by length L , adapted from Ref [9].

The detector features two anodes: The first, labelled ΔE , measures the energy loss over a sensitive length L , while the second anode, labelled E_{res} , captures the residual energy of the ions as they pass through the remaining gas. The signals from these anodes are collected and analysed separately by the data acquisition system.

Energy Loss and Signal Collection

The energy loss in the detector is described by the electronic energy loss formula, which is integrated over the length of the sensitive region to obtain the signal for ΔE :

$$\Delta E \propto \int_0^L \left(\frac{dE}{dx} \right)_e dx \quad (3.9)$$

The second anode, E_{res} , measures the energy loss over the remainder of the path as the ions travel through the gas:

$$E_{res} \propto \int_L^\infty \left(\frac{dE}{dx} \right)_e dx \quad (3.10)$$

A Frisch Grid is placed between the ionisation region and the anodes. This grid ensures that only electrons, which have greater mobility than cations, contribute to the signal, thereby reducing drift velocity effects and enhancing measurement precision.

Calibration and Optimisation

The signals collected from the anodes are recorded as channel numbers and later calibrated for known samples. To optimise detector performance, the gas pressure in the chamber is carefully controlled, as this affects both the resolution of the detector and the ion's ability to be stopped within the sensitive volume. The optimal pressure ensures the energy distribution between the two anodes is balanced for the isotope of interest.

3.2 Setups

In the study by Steier et al. (2019) [5], the overall ion detection efficiency of the degrader foil technique was found to be 15%, approximately one-third of the 45% achieved using the Si_3N_4 foil stack method. Building on these findings, we aim to assess the efficiencies of both the degrader foil and Si_3N_4 stack methods in our setup and compare their performance. We aim to

replicate similar results for the 1 MV tandem accelerator at AARAMS, as summarised in table 3.1.

Table 3.1: The overall ion detection efficiency excludes ionisation yields of the sputter source and refers to the efficiency from the anion beam to final ^{10}Be detection.

Parameter	Degrader Foil	Foil Stack
Overall ion detection efficiency	15%	45%
Stripping yield & acceler. Transm.	55%	55%
HE-transm. & detector efficiency	28%	>80%

Stacked Foil technique

The stacked foil technique is a method used to suppress isobaric interference for ^{10}Be detection. Instead of using thin foils as in the degrader technique, it employs Si_3N_4 slabs before the detector window. This setup increases the transmission of ^{10}Be cations by avoiding charge-state filtering before detection, while still effectively suppressing ^{10}B .

3.3 SRIM/TRIM

SRIM is a software package that is widely used for simulating the interaction of energetic ions with matter. SRIM includes TRIM, a Monte Carlo simulation-based code that tracks ion penetration, energy loss, and target damage in layered materials. SRIM is particularly useful for calculating ion ranges, energy deposition profiles, and sputtering yields, making it highly relevant for this thesis. The program is based on theoretical models of nuclear and electronic stopping powers and integrates experimental data where available. It is most practical for ion-solid interactions at keV to MeV energy scales [19].

3.4 Optimisation of Si₃N₄ Foil Thickness and GIC Pressure Using TRIM and Analytical Models

To optimise the detector's performance and suppress background signals from ¹⁰B, two methods were considered: An initial simulation-based approach using TRIM, and a later analytical model based on stopping range conditions, using SRIM/TRIM. Both approaches aimed to identify the optimal combination of Si₃N₄ foil thickness and detector gas pressure to allow for effective discrimination between the two ¹⁰Be and ¹⁰B at a beam energy of 1.3846 MeV, while simultaneously maximising the suppression of ¹⁰B.

Initial TRIM-Based Simulations

The first method was based entirely on TRIM simulations. In the initial step, TRIM was used to simulate the projected range distributions of ¹⁰Be and ¹⁰B in Si₃N₄ foils. An algorithm was developed to determine the optimal foil thickness that maximises transmission of ¹⁰Be ions while minimising the number of ¹⁰B ions passing through. This was done by evaluating the overlap of the two range distributions and identifying a thickness that suppresses ¹⁰B most effectively without significantly compromising the ¹⁰Be signal.

Once an optimal thickness was chosen, this value was then used in further TRIM simulations of the full detector setup. These simulations modelled the transport of ¹⁰Be ions through the Si₃N₄ foil and into a detector volume filled with C₄H₁₀ gas at varying densities. The goal was to determine the gas density range that allows valid ¹⁰Be ions to pass through the first anode and reach the second, under the assumption that we have filtered out unwanted ion species. The results were then used to estimate the optimal operating gas pressure via conversion from density.

Si₃N₄ Thickness Optimisation Algorithm Based on Range Distributions

The algorithm aims to find the optimal cutoff depth that maximises the fraction of ¹⁰Be ions that pass through the Si₃N₄ foils while minimising the fraction of ¹⁰B ions that also transmit. The core of the algorithm is the calculation of a ratio $R(D)$ for each depth D , where $R(D)$ is the ratio of the fraction of ¹⁰Be ions that traverse the foil to the fraction of ¹⁰B ions that are stopped:

$$R(D) = \frac{F_{\text{Be10}}(D)}{1 - F_{\text{B10}}(D)}.$$

The algorithm iterates over possible depths within a specified range, calculating the corresponding fractions for both ¹⁰Be and ¹⁰B ions at each depth. The goal is to select the depth where this ratio is maximised while ensuring that the fraction of ¹⁰B ions that pass through the Si₃N₄ foil does not exceed 5% (i.e., $F_{\text{B10}}(D) \leq 0.05$).

Proposed Analytical Model

This model uses range-energy relations from SRIM and stopping power approximations. A Si₃N₄ thickness will be based on the average projected range of ¹⁰Be, ensuring that most ¹⁰Be ions would exit the foil with residual energy, while a significant portion of ¹⁰B would be stopped or emerge with lower energy.

Foil Thickness Selection Strategy

To guide the selection of a suitable Si₃N₄ foil thickness, TRIM simulations were used to evaluate the projected range distributions of ¹⁰Be and ¹⁰B ions at the relevant initial energy. The goal was to choose a thickness that allows the majority of ¹⁰Be ions to penetrate the foil while stopping a substantial portion of ¹⁰B ions, or at least reducing their exit energy significantly. This balance ensures that transmitted ¹⁰Be ions retain sufficient energy to reach

the second anode of the GIC. In contrast, unwanted ^{10}B ions are either stopped in the foil or lose enough energy to be stopped in the first anode region through appropriate pressure tuning. The selected thickness of the Si_3N_4 foil will be denoted as x_μ . The respective exit energies were calculated at thickness x_μ . These exit energies were then used to calculate the pressure needed to stop ^{10}B within the first anode, while still allowing ^{10}Be to reach the second anode.

Estimation of Ion Exit Energy from SRIM Stopping Power Data

To estimate the residual (exit) energy of the ions after traversing through a Si_3N_4 foil of thickness x_μ , the stopping power data generated from SRIM were used, the stopping power as described in 2.2. Rather than assuming a constant stopping power over the ion's path, a numerical integration approach was used to accumulate energy loss over the projected range. Specifically, the total stopping power, the sum of electronic and nuclear contributions, was integrated over the depth x_μ to estimate the total energy lost:

$$\Delta E = \int_0^{x_\mu} \left(\sum_i \frac{dE_i}{dx} \right) dx = \int_0^{x_\mu} \left(\frac{dE_{elec.}}{dx}(x') + \frac{dE_{Nucl.}}{dx}(x') \right) dx'$$

The remaining energy E_{exit} is then computed by subtracting this energy loss from the known initial energy $E_0 = 1.3846 \text{ MeV}$:

$$E_{\text{exit}} = E_0 - \Delta E$$

The integration was performed numerically using the trapezoidal rule on the discrete SRIM data. The stopping power values, originally given in units of $\text{eV}/\text{\AA}$, were converted to $\text{eV}/\mu\text{m}$ to match the depth units. This method provides a more accurate estimate of ion energy at a given depth, especially where the stopping power changes significantly along the path. The exit energy of ions is directly related to their penetration depth, since energy loss is governed by the stopping power, $\frac{dE}{dx}$, at the point where they stop.

As discussed in [20, 21], to quantify the uncertainty in the final ion energy arising from the spread in penetration depth, we propagate the depth uncertainty to an energy uncertainty using a linear approximation:

$$\sigma_E = \left| \frac{dE}{dx} \right|_{\mu} \sigma_x$$

It should be noted, however, that energy deposition by a single particle is an inherently stochastic process. The Bethe-Bloch formula provides only the mean energy loss per unit length, whereas the actual distribution of energy loss follows a Landau distribution, characterised by a long tail towards higher energy losses. Therefore, the linear propagation here represents an approximation of the uncertainty due to penetration depth variation, while the intrinsic fluctuations in energy loss require a more detailed statistical treatment. Here, σ_E represents the standard deviation of the exit energy distribution, and $\left| \frac{dE}{dx} \right|_{\mu}$ is the magnitude of the stopping power evaluated at the mean range x_{μ} . This approach assumes a locally linear energy loss near the average penetration depth, which is not strictly valid in all cases. However, the assumption holds for the narrow range distributions characteristic of our ion implantation profiles.

When reporting the uncertainty in the mean exit energy, the standard error of the mean must be considered. This is given by:

$$\sigma_{E,\text{mean}} = \left| \frac{dE}{dx} \right|_{\mu} \frac{\sigma_x}{\sqrt{N}}$$

Where N is the total number of ions simulated. This expression reflects that the precision in the estimate of the average exit energy improves with increasing sample size, reducing the uncertainty by a factor of $\sqrt{N}$.

In summary, by combining the fit parameters of the depth distribution with the stopping power at the mean range, we get a statistically rigorous quantification of the uncertainty in the exit energy due to range straggling.

Foil and Gas Pressure Optimisation Procedure

To achieve separation between ^{10}B and ^{10}Be in the GIC, the strategy relies on exploiting differences in energy loss and range in Si_3N_4 and C_4H_{10} .

1. **Foil exit energies.** SRIM simulations were used to calculate the residual ion energies E_{B} and E_{Be} for ^{10}B and ^{10}Be , respectively, after passing through a Si_3N_4 foil of thickness x_μ . The initial energy of both ions is set by the accelerator and denoted $E_0 = 1.3846 \text{ MeV}$.
2. **Range in gas at standard conditions.** Using the same SRIM simulations, the projected ranges of ^{10}B and ^{10}Be ions in the detector gas were extracted at $P = 1 \text{ atm}$. These are denoted:

$$R_{\text{B}}^{(1)} = R_{\text{B}}(E_{\text{B}}, P = 1 \text{ atm}), \quad R_{\text{Be}}^{(1)} = R_{\text{Be}}(E_{\text{Be}}, P = 1 \text{ atm}).$$

The range for an ion with energy E_i can be found as the integral of the inverse stopping power over the initial energy:

$$R = \int_0^{E_i} \frac{dE}{S(E(x), \rho(P))}$$

3. **Pressure scaling.** Assuming the stopping power scales linearly with gas density (valid for low-pressure gases), the range $R(P)$ at arbitrary pressure P (in atm) is given by:

$$R(P) = \frac{R^{(1)}}{P}.$$

4. **Detector constraints with safety margins.** Let L_1 be the length of the first anode region and L_2 the length of the second. To account for tolerances and ensure reliable particle separation, we introduce small safety margins $\delta_1, \delta_2, \delta_3 > 0$. The pressure must then satisfy the following criteria:

$$R_{\text{B}}(P) = \frac{R_{\text{B}}^{(1)}}{P} < L_1 - \delta_1 \quad (\text{stop } ^{10}\text{B} \text{ safely in first anode})$$

$$R_{\text{Be}}(P) = \frac{R_{\text{Be}}^{(1)}}{P} \geq L_1 + \delta_3 \quad (\text{ensure } ^{10}\text{Be} \text{ passes through first anode})$$

$$R_{\text{Be}}(P) = \frac{R_{\text{Be}}^{(1)}}{P} \leq (L_1 + L_2) - \delta_2 \quad (\text{stop } ^{10}\text{Be} \text{ safely within second anode})$$

Solving these inequalities yields a conservative pressure window:

$$\max \left(\frac{R_{\text{B}}^{(1)}}{L_1 - \delta_1}, \quad \frac{R_{\text{Be}}^{(1)}}{(L_1 + L_2) - \delta_2} \right) \leq p \leq \frac{R_{\text{Be}}^{(1)}}{L_1 + \delta_3}.$$

If this interval is non-empty, a suitable pressure exists to stop ¹⁰B in the first anode while allowing ¹⁰Be to reach (but not exceed) the second. The safety margins ensure robust performance despite uncertainties in energy, pressure stability, or detector geometry.

3.5 Statistical Approach for Clustering and Outlier Detection

Since raw ion count data contains noise and potential outliers, an optimised clustering approach was developed to accurately isolate ^{10}Be signals and minimise false detections. Traditionally, regions of interest (ROIs) in such datasets have been selected manually, often by hand-drawing boundaries around suspected signal clusters. However, this approach introduces subjectivity, lacks reproducibility, and may not generalise well across datasets with varying characteristics. A custom statistical method, Optimised KNN with Outlier Handling (OKOH), was developed to address these limitations. The process begins with parameter optimisation for KNN-based clustering, using silhouette scoring and the elbow method to identify the optimal number of neighbours and cluster configurations. Once an initial clustering is achieved, Local Outlier Factor (LOF) is applied to detect and exclude data points that significantly deviate from the local density, thereby removing potential noise and edge anomalies. A final re-clustering step is performed on the refined dataset to delineate a clean and accurate ROI for the ^{10}Be signal. This method offers a reproducible, objective, and data-driven alternative to manual ROI selection, improving the reliability and transparency of signal classification in complex datasets.

K-Means Clustering for Data Grouping

The dataset was first subjected to K-Means clustering to identify natural groupings within the data. The K-Means algorithm is a partitioning method that iteratively assigns each data point to one of the k clusters by minimising the within-cluster variance. Two evaluation metrics were used to determine the optimal number of clusters (k): The silhouette score and the elbow method. The silhouette score, which measures cohesion and separation of clusters, was calculated for each value of k within a predefined range.

Scores range from -1 to $+1$, with higher values indicating better-defined and well-separated clusters. In parallel, the elbow method was used to plot the total within-cluster sum of squares (WCSS) against different k -values. The point at which the rate of improvement sharply decreases, forming an elbow, was identified as a candidate for the optimal cluster number. The final value of k was selected based on a balance of these two criteria. This optimisation should ensure the clustering step is robust, reproducible, and well-aligned with the dataset's underlying structure.

Outlier Detection and Filtering

Following the initial clustering, a critical refinement step was undertaken to filter out outliers that could affect the precision and reliability of the ^{10}Be counts. Outliers were detected using the Local Outlier Factor (LOF) algorithm, which evaluates the local density of data points and identifies less dense points compared to their neighbours [22]. The LOF algorithm operates as follows:

1. **Reachability Distance:** For a point A , the reachability distance to a neighbour B is defined as:

$$\text{reachability-distance}_k(A, B) = \max\{k\text{-distance}(B), d(A, B)\},$$

where $k\text{-distance}(B)$ is the distance to the k -th nearest neighbor of B , and $d(A, B)$ is the Euclidean distance between A and B .

2. **Local Reachability Density (LRD):** The LRD of A is the inverse of the average reachability distance from A to its k nearest neighbours:

$$\text{lrd}_k(A) = \frac{|N_k(A)|}{\sum_{B \in N_k(A)} \text{reachability-distance}_k(A, B)},$$

where $N_k(A)$ is the set of k nearest neighbours of A .

3. **LOF Score:** The LOF score for A is computed as:

$$\text{LOF}_k(A) = \frac{1}{|N_k(A)| \text{lrd}_k(A)} \sum_{B \in N_k(A)} \text{lrd}_k(B).$$

A score close to 1 indicates that A has a similar density to its neighbours, while $\text{LOF}_k(A) \gg 1$ suggests an outlier. Points with $\text{LOF}_k(A) > \text{threshold}$ were flagged as outliers and removed from the dataset.

Re-clustering of the ROI

To further refine the grouping of the ROI cluster, the filtered data points were re-clustered using K-Means with a reduced number of clusters. This step ensured that the ROI cluster remained well-defined and homogeneous after removing outliers.

The same algorithm is used for Clustering, Outlier Detection, and Ion Counting to ensure consistency and prevent bias when counting ions.

3.6 Counting of ^{10}Be at AARAMS

The SO110 ion source sample wheel can hold up to 50 samples, each undergoing multiple measurements called runs. Each run comprises several measurement cycles referred to as blocks. The number of blocks per run is determined by stop conditions set in the Tandetron control software via a batch file. These conditions may be based on either a fixed number of blocks or a target count of ^{10}Be . The selection of stop conditions hinges on the required sample precision. Data analysis involves aggregating blocks into runs and averaging the runs to determine sample means, reported as the final results of cosmogenic isotope analysis. This methodology adheres to the approach outlined in [23], where further details can be found.

Blocks

A block represents a single measurement cycle where the cation is recorded. The rare isotope ratio for a block is given by

$$R_{\text{Block}} = \frac{neq}{It}$$

where n is the isotope count within the ROI, e is the elementary charge, q is the ion charge state, I is the stable isotope current (measured using FCs), and t is the detector live time. The uncertainty in block measurements is calculated as

$$\sigma_{R_{\text{Block}}} = \frac{eq}{It} \sqrt{n}$$

Runs

A run consists of multiple blocks for a single sample. The AMS software stop conditions determine the number of blocks per run. The rare isotope ratio for a run is computed as a weighted average of block ratios:

$$R_{\text{run}} = \frac{\sum_i R_{\text{block},i} I_i t_i}{\sum_i I_i t_i}$$

where $I_i t_i$ represents the stable isotope charge as the weighting factor. The error in the weighted average of block ratios is given by the maximum of the counting statistical error and the variance-based error:

$$\sigma_{R_{\text{Run}}} = \max \left[\sigma_{R_{\text{Run}}}^{\text{count}}, \sigma_{R_{\text{Run}}}^{\text{var}} \right] = \max \left[\frac{\sum_i \sigma_{R_{10, \text{block}, i}} I_i t_i}{\sum_i I_i t_i \sqrt{n_{\text{block}}}}, \frac{\text{stdev} \langle R_{\text{block}} \rangle}{\sqrt{N_{\text{block}}}} \right]$$

Samples

A sample measurement consists of multiple runs, and the sample ratio is computed as the weighted average of run ratios:

$$R_{\text{sample}} = \text{wavg} \langle R_{\text{run}, x} \rangle$$

where the error is the maximum of the counting statistical error and the variance-based error:

$$\sigma_{R_{\text{Sample}}} = \max \left[\sigma_{R_{\text{Sample}}}^{\text{count}}, \sigma_{R_{\text{Sample}}}^{\text{var}} \right] = \max \left[\text{SE} \langle \sigma_{R_{\text{run}, x}} \rangle, \frac{\text{stdev} \langle R_{\text{run}, x} \rangle}{\sqrt{n_{\text{run}, x}}}} \right]$$

The distinction between counting statistics and sample variance applies strictly speaking when summarising blocks into runs. When aggregating runs into sample means, the underlying runs may include sample variance rather than counting statistics.

Chapter 4

Analysis and Discussion

This chapter presents SRIM/TRIM simulation results for optimising stacked foil thickness and gas density of the GIC for the initial and proposed methods for the implementation of the stacked-foil technique. Additionally, includes results from the Degraded and Stacked foil techniques and discusses their outcomes and challenges.

4.1 Initial Method: TRIM Simulations and Parameter Optimisation

TRIM simulations were conducted at various ion energies to evaluate their effect on the optimal thickness of the Si_3N_4 foil. This investigation was initially motivated by the hypothesis that energy-dependent stopping powers could enhance separation efficiency. While ^{10}B loses energy more rapidly than ^{10}Be in both Si_3N_4 and C_4H_{10} due to its higher nuclear charge ($Z = 5$ vs. $Z = 4$), the relative difference in their stopping powers is smaller in C_4H_{10} than in Si_3N_4 . This material-dependent scaling was expected to produce favourable separation behaviour when combining both materials in the detection system. However, subsequent discussions and observations indicated that the energy ranges explored were not well-matched to the actual operational conditions.

Si₃N₄ Thickness Optimisation

To determine this optimal thickness, TRIM simulations were conducted for both ¹⁰B and ¹⁰Be. The Si₃N₄ foil was modelled as a single layer with a density of $\rho(\text{Si}_3\text{N}_4) = 9.525 \times 10^{22} \text{ atoms/cm}^3$ and a thickness of $3 \mu\text{m}$. The atomic composition was Si 42.8% (60.0 mass%) and N 57.1% (39.9 mass%). The energy of the ions was set to 1.3846 MeV, derived from the 1 MV terminal voltage, as detailed in eq. (2.6).

Energy Dependence and Isotopic Ratio Immunity

To account for uncertainties in the thicknesses of the Si₃N₄ foil, we conducted simulations to explore the energy dependence of the penetration depth for both ¹⁰B and ¹⁰Be.

We investigated the effect of varying ¹⁰B/¹⁰Be ratios on the optimisation results. While a 1:1 ratio was used for most comparative simulations to simplify analysis, additional simulations were conducted using more experimentally realistic ratios ranging from 1:10 to 1:10,000. Across this range, at set energies ranging from 900 keV to 1.4 MeV, the optimal cutoff depth varied minimally, demonstrating some immunity to changes in background levels that can alter the ratio. This also ensured that we don't have to be too worried about fluctuations in samples, as they will have different abundances of ¹⁰B.

In all cases, a foil thickness of $1.86 \mu\text{m}$ was sufficient to completely stop ¹⁰B, including its low-intensity tail, indicating that background contamination at this thickness is negligible.

Figure 4.1 shows simulations over an energy range of 1000 keV to 1.4 MeV demonstrating that the optimal foil thickness increases linearly with the ion energy. It is shown with a best-fit model given by $y = ax + b$, where x is the optimal cutoff depth as calculated by the ratio in section 3.4 and y is the simulated ion energy, the graph displays ion energy on the x-axis and depth on the y-axis for clarity. We observe that the data points deviate

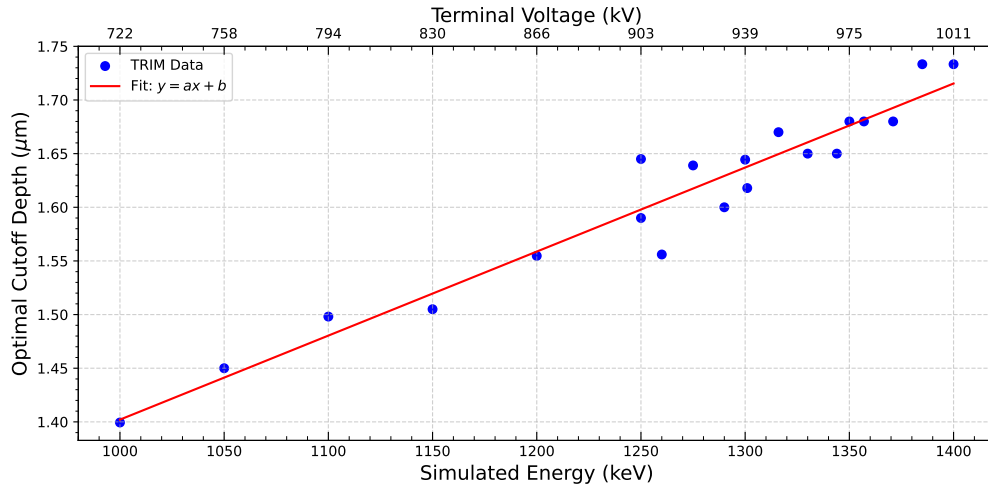


Figure 4.1: Simulated optimal cutoff depth using Si_3N_4 in the energy range 1000 keV to 1400 keV. The linear fit yields $a = 0.00078 \pm 0.00005 \mu\text{m}/\text{keV}$ and $b = 0.62 \pm 0.059 \mu\text{m}$.

more from the linear fit beyond 1200 keV. This is likely due to the reduced number of simulated ions; only 100 ions were used for these higher-energy points, compared to 1000 ions for the lower-energy range. The smaller sample size results in greater statistical uncertainty and increased variability. Nevertheless, the overall trend still exhibits linear behaviour.

Determining Foil Thickness at 1.3846 MeV

Using a 1:1 $^{10}\text{B}/^{10}\text{Be}$ ratio, we evaluated the depth distribution of both ions at 1.3846 MeV. As shown in fig. 4.2, the two distributions are visible. Both ion depth profiles exhibit a noticeable left skew, with tails extending toward shallower depths. This asymmetry indicates that while most ions penetrate to within the projected range, a fraction scatters back or stops earlier, causing the distributions to deviate from an ideal Gaussian shape.

This skewness arises because the probability distribution of energy loss, dE/dx , is governed by the Landau distribution [24].

Transforming energy loss into range, $R = 1/S = 1/(dE/dx)$, effectively inverts the variable. Under this transformation, Landau's characteristic

positive skew (long right tail due to rare large energy losses) becomes a negative skew (long left tail in range). Physically, this can be interpreted as hard collisions leading to rapid stopping.

Mathematically, if $Y = g(X)$ is a one-to-one (invertible) transformation and X has probability density function (PDF) $f_X(x)$, then the PDF of Y is given by; $f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$ [25]. In this case, with $g(x) = 1/S$, the transformation introduces asymmetry in the opposite direction, consistent with the observed negative skew in the range distribution fig. 4.2.

While the maximum penetration depth of ^{10}B could serve as a strict cutoff, this would exclude part of the ^{10}Be tail and reduce overall transmission efficiency. Therefore, the foil thickness of $1.86\ \mu\text{m}$ was not considered suitable for this initial method.

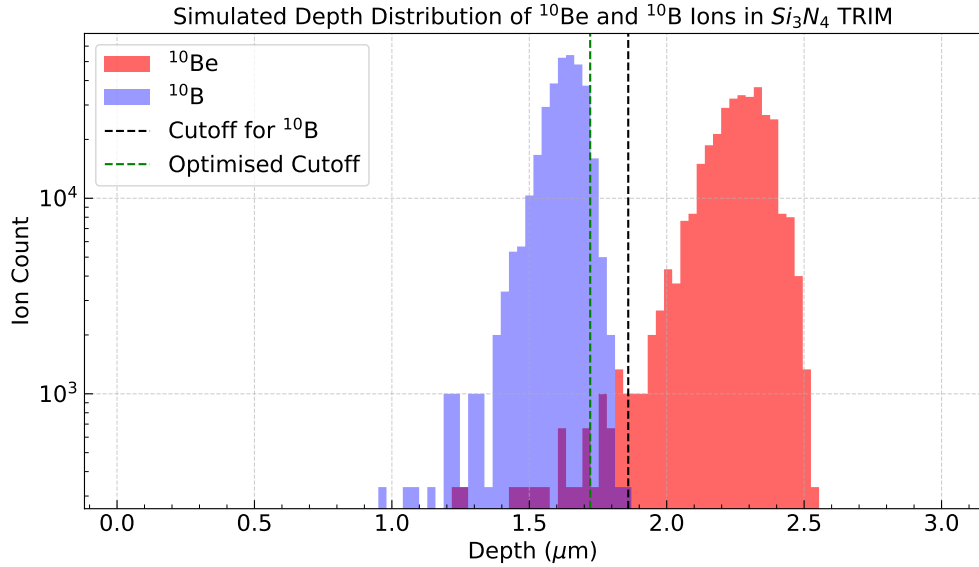


Figure 4.2: Simulated depth distributions of ^{10}B (blue) and ^{10}Be (red) in Si_3N_4 . The black dashed line indicates the maximum depth of ^{10}B , while the green dashed line marks the optimised cutoff depth ($1.721539\ \mu\text{m}$), selected to maximise ^{10}Be transmission while minimizing ^{10}B penetration.

The optimal cutoff depth was determined to be $1.767 \pm 0.005\ \mu\text{m}$, at which point 98.70% of ^{10}Be ions penetrate the foil, compared to just 7.10%

of ^{10}B . Of the 352,667 particles reaching beyond this depth, 93.29% are ^{10}Be and only 6.71% are ^{10}B , ensuring effective isotopic suppression.

The uncertainty in the optimal cutoff depth was estimated as the standard deviation of the mean, derived from the distribution of optimal values across 50 simulation runs.

We selected a practical thickness of $1.75\ \mu\text{m}$ for the Si_3N_4 foil, corresponding to commercially available values. This deviation is slight but remains within acceptable tolerance limits. The expected change in separation performance is negligible and still satisfies the $\geq 95\%$ ^{10}Be transmission criterion.

Gas Density Optimisation

A model of the detector is shown in fig. 4.3, where the simulated trajectories of ^{10}Be ions are illustrated by colourful lines.

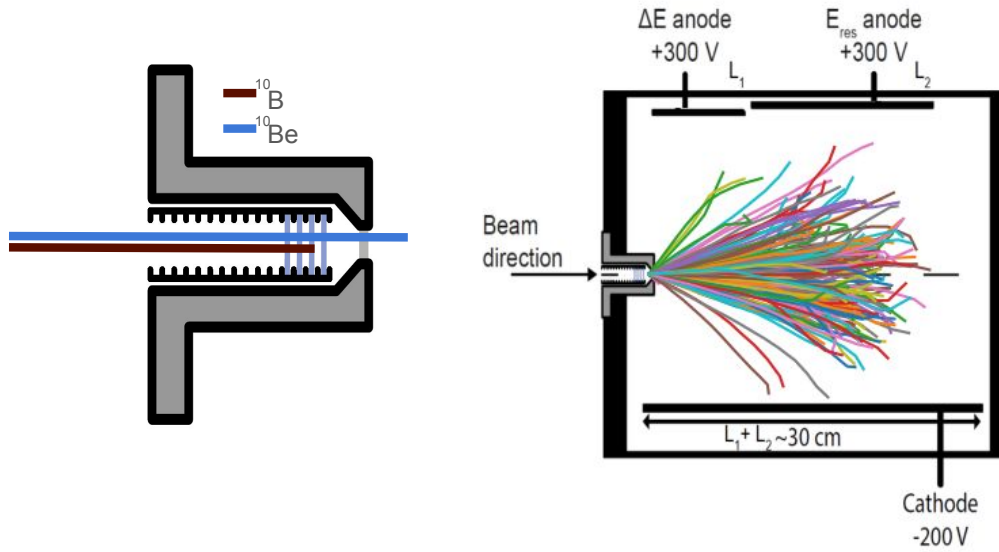


Figure 4.3: Left: Schematic of the foil holder for the detector, focusing on ^{10}Be . Right: Modelled schematic of the detector setup showing simulated trajectories of ^{10}Be ions represented by colourful lines.

The ^{10}Be ions enter the detector after passing first through a Si_3N_4 foil of thickness $1.75\ \mu\text{m}$ with an initial energy of 1.3846 MeV. After the foil,

ions travel through the C_4H_{10} gas, modelled with a composition of 80.0 atomic % C (82.7 mass%) and 20.0 atomic % H (17.3 mass%), with density varied in simulations between 3×10^{-6} and $8 \times 10^{-6} \text{ g/cm}^3$.

We simulated ion trajectories in increments of 10,000 eV inside the detector volume to analyse how gas density influences the detection efficiency of valid ions, if they satisfied some boundary conditions.

Counting detections in the simulated detector required two conditions: A particle must pass the first anode, which spans roughly one-third of the effective detector length $L_{\text{effective}} \approx 30 \text{ cm}$, and it must reach the second anode without exceeding $L_{\text{effective}}$. It is important to note that the simulation is two-dimensional, considering only vertical and horizontal movements, so particles deviating significantly out of plane would not be accounted for accurately.

The resulting counts of valid ^{10}Be ions as a function of gas density are shown in fig. 4.4. Valid counts increase steeply up to a gas density corresponding to approximately 1.2–1.7 mbar, then plateau before falling off sharply outside this range. The average ion path length remains relatively stable within this pressure window.

The uncertainty in the number of valid ions was calculated by performing simulations of 250 ions, where each ion was independently simulated 10 times. For every individual simulation (i.e., for each ion i in repetition j), approximately 108 steps were recorded, from which a mean step count μ_{ij} and standard deviation σ_{ij} were calculated. The overall average step count was then computed as

$$\bar{\mu} = \frac{1}{N_{\text{ions}} N_{\text{reps}}} \sum_{i=1}^{N_{\text{ions}}} \sum_{j=1}^{N_{\text{reps}}} \mu_{ij}. \quad (4.1)$$

The final uncertainty was determined as the standard error of this distribution, given by $\text{Uncertainty} = \frac{\sigma(\{\mu_{ij}\})}{\sqrt{N_{\text{ions}} N_{\text{reps}}}}$, where $\sigma(\{\mu_{ij}\})$ is the standard deviation of all mean step counts μ_{ij} .

For the ion range, defined as the difference between the maximum and minimum depth positions for each ion trajectory, the uncertainty in the

average range was computed analogously. The range value for ion i in simulation j , denoted ρ_{ij} , was calculated across all valid ions and repetitions. The overall average ion range was calculated as

$$\bar{\rho} = \frac{1}{N_{\text{ions}}N_{\text{reps}}} \sum_{i=1}^{N_{\text{ions}}} \sum_{j=1}^{N_{\text{reps}}} \rho_{ij}. \quad (4.2)$$

The uncertainty in the average ion range was obtained as the standard error of these values: $\text{Uncertainty}_{\text{range}} = \frac{\sigma(\{\rho_{ij}\})}{\sqrt{N_{\text{ions}}N_{\text{reps}}}}$, where $\sigma(\{\rho_{ij}\})$ is the standard deviation of the ion range means ρ_{ij} across all ions and repetitions.

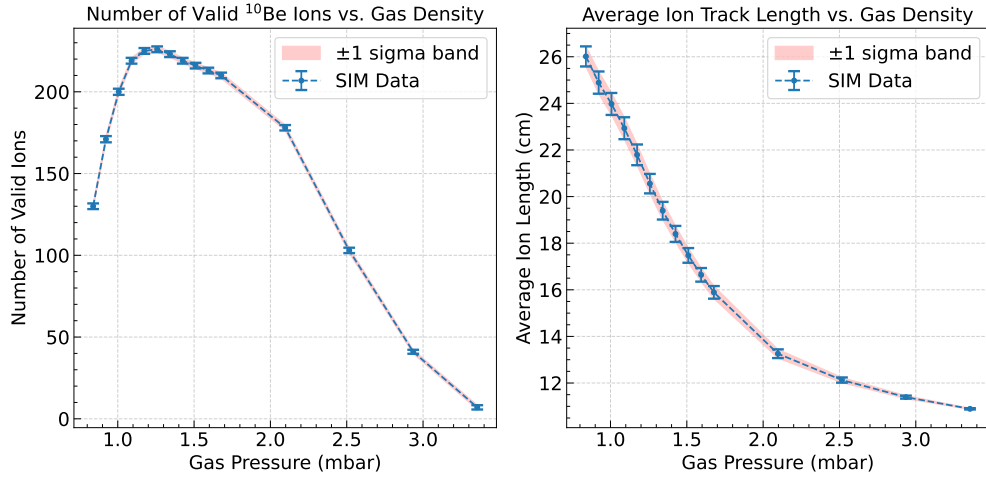


Figure 4.4: Left: Valid ^{10}Be ion counts at varying gas densities, illustrating the optimal gas density range for detection efficiency. Right: Average ion trajectory lengths for valid ions at different gas densities

In the lower pressure regime, we had concerns regarding the accuracy of the pressure gauge used in the detector, as its readings at very low pressures might not be reliable, so we initially discarded any ideas of operating at pressures below the peak in fig. 4.4 (left) as we thought it would be of no use.

The curve in fig. 4.4 (left) resembles a broad, asymmetric parabola, with the valid ion count peaking within a narrow pressure range. This trend aligns with expectations based on the boundary conditions defined for valid ion detection. At very low pressures, the gas density is too low to sufficiently

slow the ions, causing them to overshoot the detector, contributing to the observed asymmetry. At high pressures, although not a concern here, the increased stopping power would cause ions to lose energy too quickly and stop short of the detector. The peak region represents an optimal balance, where ^{10}Be ions lose just enough energy to be fully stopped within the detector's active volume.

The maximum ion energy of 1.3846 MeV imposes constraints on the gas pressure range. Increasing pressure excessively leads to greater ion energy loss, which can prevent detection. Since the beam energy cannot be increased beyond this limit, and the foil thickness is optimised to stop most ^{10}B ions, the operational window for gas pressure is tightly constrained.

This narrow optimal pressure range raised some concerns regarding the pressure gauge and its accuracy.

At the time, we considered a new approach to address concerns regarding the low pressure. Reducing the foil thickness would allow more ions to pass through, potentially increasing detection rates, though at the risk of saturating or even damaging the detector. However, pushing the pressure too high would cause excessive energy loss, preventing ions from reaching the anodes. Despite the uncertainty in low-pressure measurements and the constraint of a fixed ion energy, this strategy seemed like a reasonable and practical direction at the time.

4.2 Proposed Method: Analytical Model and Pressure Regime Analysis

Si_3N_4 Thickness Determination via Ion Distributions

Using simulated data on ion transport in matter, we can assess the projected range distributions of both ^{10}Be and ^{10}B ions at an initial energy of 1.3846 MeV in Si_3N_4 .

Figure 4.5 shows the distribution of ^{10}Be and ^{10}B penetration depths with Gaussian fits over the denser regions (DR). By analysing and fitting with a Gaussian profile, we get a mean range (x_μ^{DR}) with a corresponding standard deviation (σ_x) for the two DR's of the respected ions in Si_3N_4 . It is the $x_\mu^{DR}(^{10}\text{B})$, mean range of DR(^{10}B), that we are interested in getting as we want to compare this value with the overall average projected range of ^{10}Be that includes the lower tail. The mean DR penetration depth of ^{10}Be in

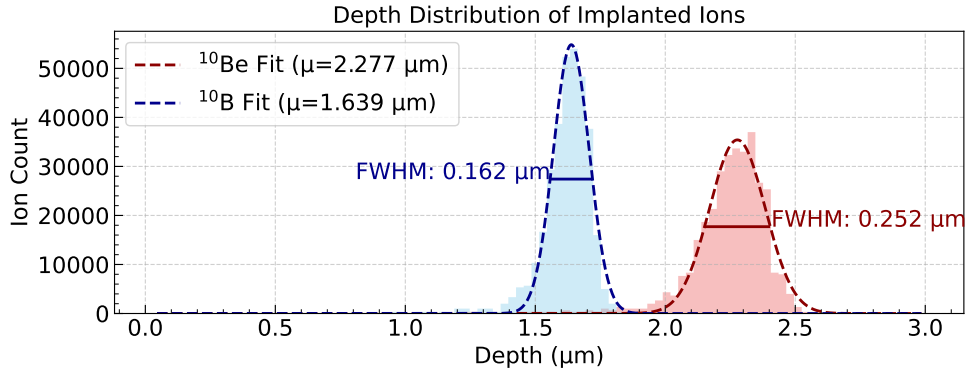


Figure 4.5: Depth distribution of ^{10}Be and ^{10}B in Si_3N_4 , characterized by Gaussian profiles.

Si_3N_4 was found to be approximately $x_\mu^{DR}(^{10}\text{Be}) = 2.28 \mu\text{m}$, with a FWHM of $0.252 \mu\text{m}$. The mean DR penetration depth $x_\mu^{DR}(^{10}\text{B}) = 1.639 \mu\text{m}$, with a FWHM of $0.162 \mu\text{m}$ indicating a relatively more narrow implantation profile. Due to the noticeable left skew and tail extending toward shallower depths, the overall average projected range of ^{10}Be was found to be $x_\mu(^{10}\text{Be}) =$

$1.515 \pm 0.005 \mu\text{m}$. Although the distribution exhibits a slight low-range tail, the majority of ^{10}Be ions penetrate the Si_3N_4 foil deeper than ^{10}B .

In comparison, the ^{10}B distribution is centred at a shallower depth of $1.64 \mu\text{m}$, with a FWHM of $0.16 \mu\text{m}$, showing that a substantial portion of these ions are stopped within the foil before exceeding the lower tail of the ^{10}Be distribution. Nevertheless, some ^{10}B ions still transmit through with reduced energy, due to the Landau distribution.

These results correspond to a 1:1 ion ratio, but further analysis with different ratios indicated that a more suitable foil thickness would be around $1.5 \mu\text{m}$, which can be ordered. The Si_3N_4 thickness was reduced, which also increases the likelihood that ^{10}Be ions emerging on the other side of the foil retain higher energy, thus improving their chances of being detected.

Post-Foil Energy Estimation via Range-Stopping Power Analysis

With the Si_3N_4 foil thickness fixed at $1.515 \pm 0.005 \mu\text{m}$, the next step was to estimate the energy of the ions after traversing the Si_3N_4 foil. This was done by comparing the projected range distribution to the stopping power curve. This method allows us to estimate the exit energies of both ^{10}Be and ^{10}B ions at the selected foil thickness. To verify the reliability of this approach, we compared the tabulated SRIM stopping powers with a numerical derivative of the SRIM-generated energy-range data. The close agreement between these two methods, shown in fig. 4.6 (top), supports the use of the numerical method for interpolating the exit energies for a given penetration depth, as it might not have been specifically simulated. As we approach the end of the stopping power curves for the ions, the trapezoidal numerical method begins to produce a poorer fit because the gradient approach is less accurate at boundaries. However, this has negligible effects since the deviations are minimal compared to the entire curve. The exit energies of the ions were calculated to $E_{^{10}\text{Be}}(x_\mu) = 743.6 \pm 0.2 \text{ keV}$ and $E_{^{10}\text{B}}(x_\mu) = 245.7 \pm 0.2 \text{ keV}$ and can be extracted from fig. 4.7.

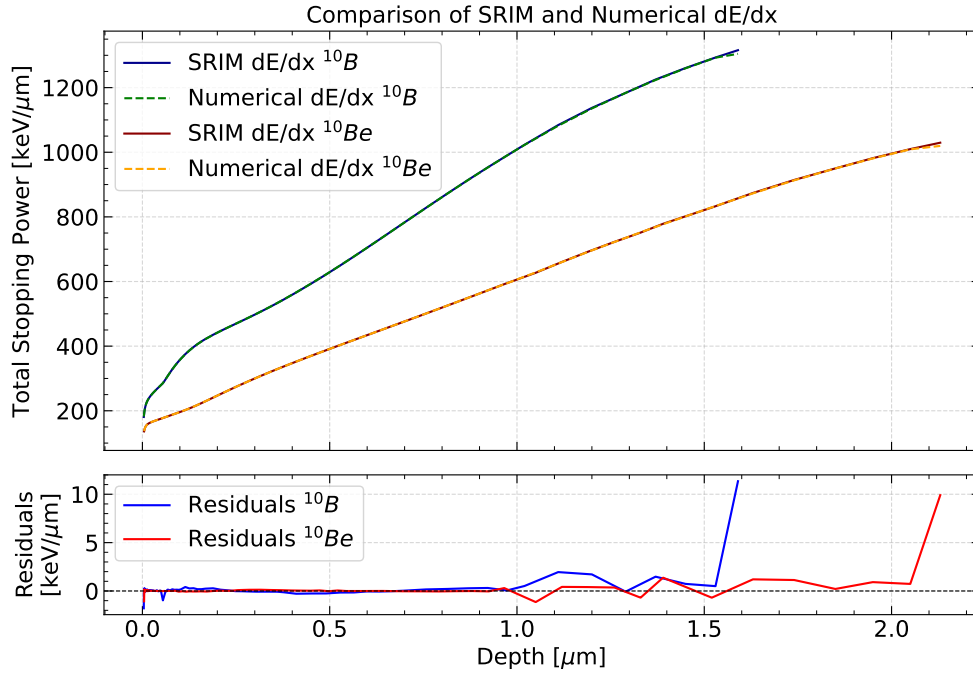


Figure 4.6: Comparison of SRIM stopping powers (solid line) and numerical derivatives of SRIM energy-range data (dashed line) for ^{10}Be and ^{10}B ions in Si_3N_4 (Top). These two curves show close agreement as shown in the residual plot (Bottom).

GIC Pressure Range Proposal

Figure 4.8 shows the valid operating pressure windows as a function of foil thickness. These pressure ranges are determined based on the exit energies of the ions and their corresponding ranges in the detector gas. The data span foil thicknesses from $1.3\ \mu\text{m}$ to $1.6\ \mu\text{m}$, with the pressure windows narrowing as the foil thickness decreases. This analysis was conducted to identify any trends.

From the data, we observe that thinner Si_3N_4 foils generally correspond to higher pressure regimes and exhibit lower uncertainty. There appears to be a nearly linear trend with a negative slope. Interestingly, at thicknesses of $1.55\ \mu\text{m}$ and $1.6\ \mu\text{m}$, the pressure ranges are nearly identical, an outcome that was not initially expected. This could be attributed to specific set

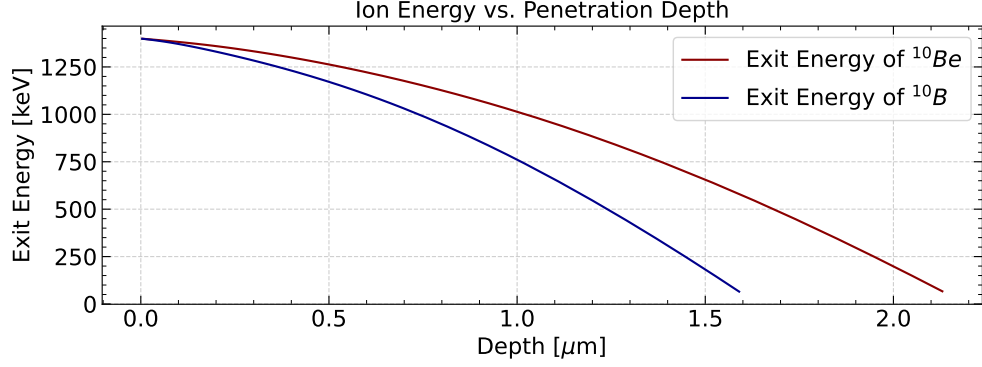


Figure 4.7: Estimated exit energies of ^{10}Be and ^{10}B ions after traversing Si_3N_4 foil can be read from the graphs, derived from range-stopping power analysis.

conditions or the possibility that the energy of ^{10}Be ions at these thicknesses is too low to yield a distinguishable difference with the boundaries we have for pressure calculations - this was not inspected any further.

For instance, at a foil thickness of $1.51\ \mu\text{m}$, the operating pressure ranges from 6.75 to 17.65 mbar, corresponding to ion energies of 743.60 keV for ^{10}Be and 245.70 keV for ^{10}B . This specific pressure regime is selected to maintain sufficient energy in the ^{10}Be ions, enough for reliable detection, while simultaneously reducing the energy of the ^{10}B ions to a level where they are less likely to pass through the detector and oversaturate it.

Each foil thickness is associated with an average pressure value, taken from its respective window, to make comparison and visualisation easier. The lower and upper bounds of the pressure window are treated as uncertainties and represented as error bars. This implies that, for ^{10}Be detection at a Si_3N_4 foil thickness of $1.5\ \mu\text{m}$ and an accelerator energy of 1 MV, the optimal operating pressure should be $P = 12.2 \pm 5.45\ \text{mbar}$

The uncertainty in foil thickness is estimated to be 1% of the nominal thickness value. This estimate arises from the trapezoidal numerical integration method used to approximate energy loss curves as a function of thickness, which introduces inherent uncertainties. This estimate is meant to take account of the residuals observed between the SRIM data and the

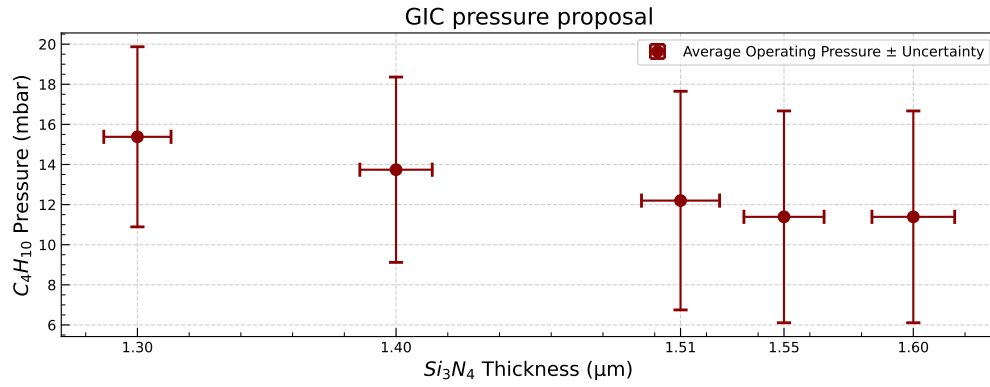


Figure 4.8: Valid operating pressure windows as a function of foil thickness. Markers indicate average pressures, with error bars representing the range bounds.

trapezoidal approximation, which become more significant for thicknesses exceeding 1 μm see fig. 4.6 (bottom).

4.3 Isotopic Ratio Efficiency Using Degraded Foils

Identifying Isotopes

The spectrum obtained using the degraded foil method is shown on fig. 4.9 (centre). It is clear to see that there are two distinctive clusters in the lower left corner. There is also an elongated shape, which is believed to be a result of coincidences from the two clusters, as well as ion energy straggling in the degraded foil. Additionally, some scattering effects can contribute to the elongated shape.

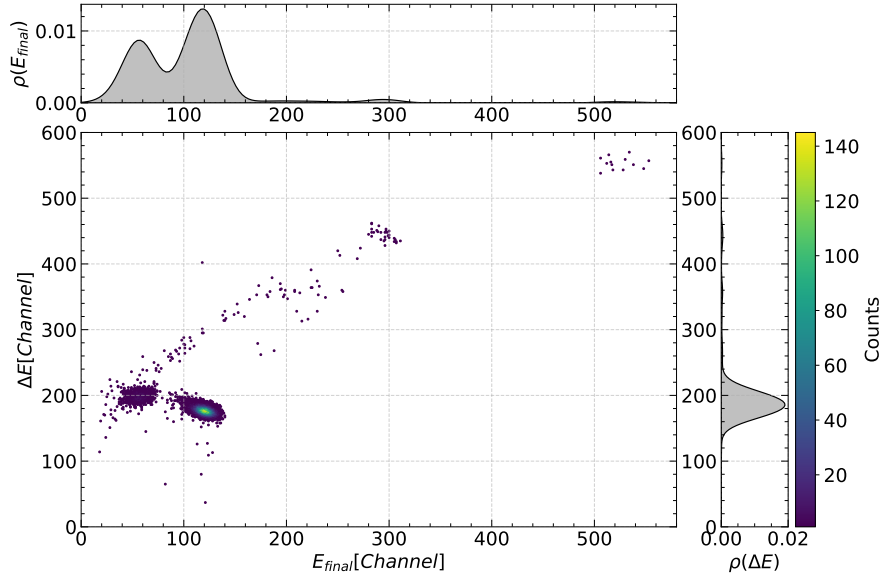


Figure 4.9: Spectrum of a $^{16}\text{O}^{10}\text{Be}$ sample (Be51 appendix A) using degraded foil method is showcased in the centre. The top figure showcases the distribution of ions of the E_{final} anode. The right figure showcases the distribution of ions for the ΔE anode. Corresponding to anodes depicted in fig. 3.7.

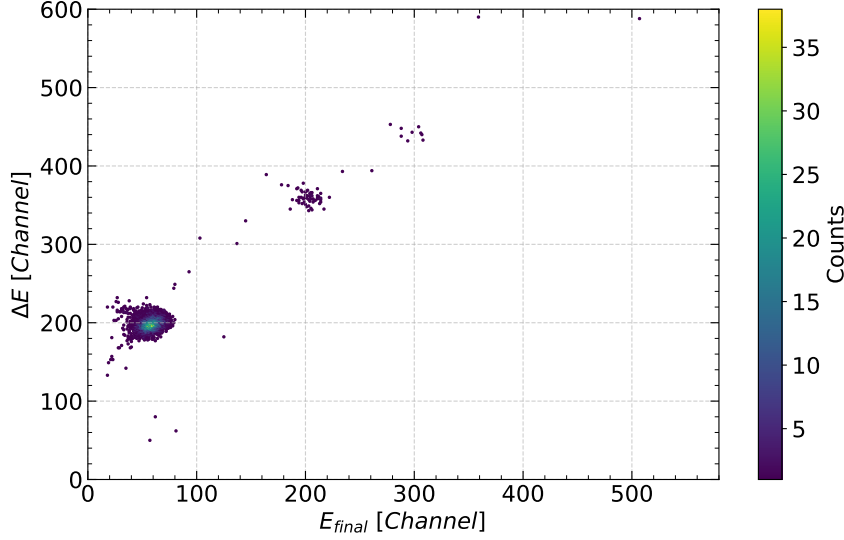


Figure 4.10: Spectrum from a blank sample (appendix A) used to identify the background contribution of ^{10}B . This allows for the isolation of the ^{10}B cluster and, by comparison with the spectrum from the $^{10}\text{Be}^{16}\text{O}$ sample, the identification of the ^{10}Be peak.

It is reasonable to expect that since ^{10}B and ^{10}Be , if both originating from the beam, should have the same charge state, and belonging to the same isospin multiplet that they will be close lying neighbours in the measured spectrum, this assumption is validated in fig. 4.9. We can identify the ^{10}B ions by analysing spectra of ^{10}B samples as well as blank samples to identify the background [26, 27]. By doing this, we can distinguish between what cluster belongs to ^{10}B and what cluster belongs to ^{10}Be .

We can reliably identify ^{10}B using blank samples, as illustrated in fig. 4.10, where a dominant cluster appears with minor scattering across the plot. In addition, two smaller diagonal clusters, believed to be coincidence events, are visible, likely resulting from particles depositing energy in a correlated manner along the same trajectory. The nearby peak, which only appears when using $^{10}\text{Be}^{16}\text{O}$ samples (as shown in fig. 4.9), can therefore be reasonably attributed to ^{10}Be .

The possibility of interference of ^{10}B counts migrating over to ^{10}Be as

we measure for longer times is negligible, as shown in fig. 4.11, when using the degrader method.

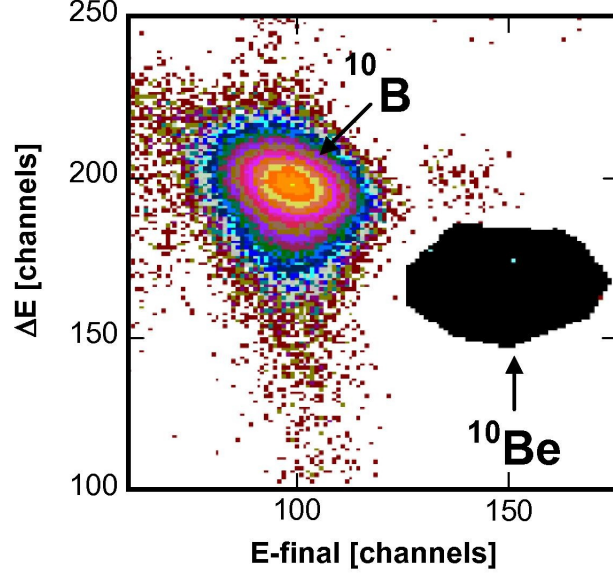


Figure 4.11: Spectrum of a deliberately exaggerated ^{10}B beam, illustrating the negligible interference in the ^{10}Be window (shown in black). Detector suppression is approximately 5 to 6 orders of magnitude. The figure is adapted from [26].

The distinctiveness of these two ions in the spectra can also be seen in the distribution, $\rho(E_{final})$, in fig. 4.9. This distribution (top box of fig. 4.9) shows two distinct peaks, with only minor overlap. The left peak is identified as ^{10}B based on reference spectra from blank samples and ^{10}B samples.

Clustering groups using OKOH

We began by optimising the parameters of the OKOH algorithm to ensure robust clustering performance. With these optimised settings in place, we applied the algorithm to the dataset, which yielded an effective number of clusters equal to 4. Two additional clustering validation techniques further supported this result. Notably, the Elbow Method showed a clear inflexion point at $k = 4$, as illustrated in fig. 4.12, where the reduction in inertia began to level off, indicating that four clusters represented an optimal balance between model complexity and explanatory power.

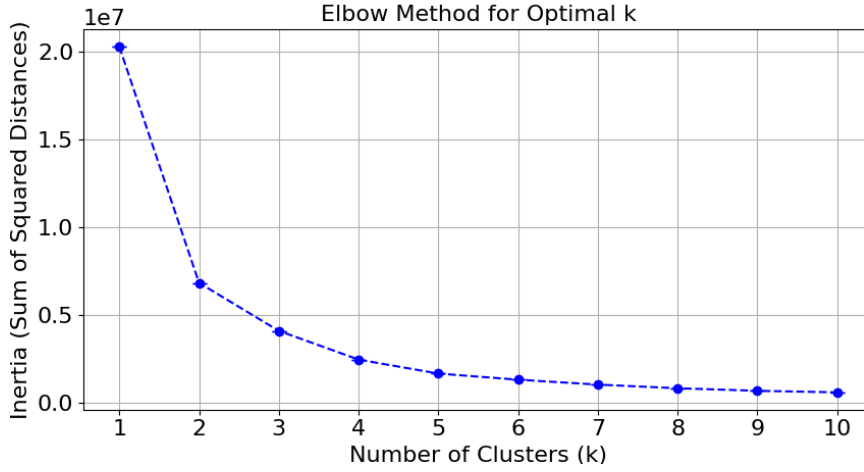


Figure 4.12: Elbow Method for determining the optimal number of clusters. For each k value, the clustering was run 100 times, and the inertia was averaged over multiple runs. The optimal $k = 4$ is identified at the inflexion point, where the reduction in inertia begins to slow.

Additionally, silhouette analysis, which evaluates the cohesion within clusters and their separation from one another, confirmed that the clustering was well-defined, with a mean silhouette score of 0.78 across all clusters. The four clusters identified by the K-Means algorithm are illustrated as shaded regions in fig. 4.13. Cluster 2, which we attribute to ^{10}Be ions, achieved the high silhouette score of 0.726, indicating strong internal cohesion and clear separation from other clusters. This suggests that, under the general degrader foil method, the ^{10}Be cluster is statistically distinguishable from ^{10}B and other ion species. Cluster 0, which corresponds to ^{10}B ions, had a silhouette score of 0.698, reflecting good clustering quality, though with slightly less distinct separation. The lower score can be attributed to greater spread within the cluster, a characteristic also observed in pure ^{10}B samples. Cluster 3, the elongated cluster, showed moderate clustering performance with a score of 0.663, indicating some degree of overlap with neighbouring clusters. Cluster 1 had the highest silhouette score of 0.925, suggesting strong cohesion. Since clusters 1 and 3 are not relevant to our analysis, they will not be discussed further.

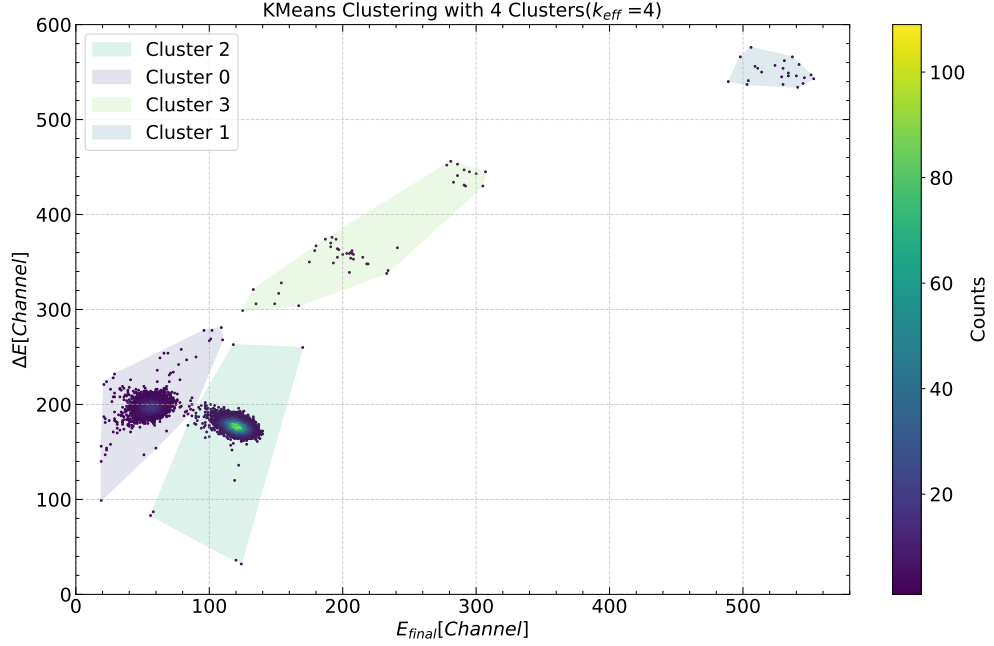


Figure 4.13: KNN clustering of the spectrum from $^{16}\text{O}^{10}\text{Be}$ samples using the degraded foil method. This spectrum was selected for its visual clarity and appeal. See appendix B (Degraded method) for other spectra.

Though the silhouette score for the cluster containing ^{10}Be did not reach the ideal value of 1, the result was still informative, as the score of 0.726 indicated a high degree of cohesion within the cluster and strong separation from others. This level of distinction supports the conclusion that ^{10}Be ions form a well-defined group, making the identification and isolation of these ions reliable for further analysis. Notably, there were some points within this cluster that did not belong, with the most prominent ones located near the four corners of the cluster. These points are believed not to be associated with the ^{10}Be ions. The LOF component of the OKOH algorithm identified these outliers, setting an outlier score cutoff at -1.333, with the assumption that 5% of the data would be outliers, corresponding to the 5th percentile of the LOF score distribution.

After filtering the outliers, the cohesion of the remaining points in the ^{10}Be cluster improved noticeably. The new silhouette score for the filtered

ROI cluster was 0.424, which, while lower than the score of 0.726, indicates reasonable cohesion. This decrease in the silhouette score compared to the initial clustering suggests that, while the filtered data is more cohesive, some overlap or ambiguity may require further refinement.

A key observation was the mean distance to the centroid and its variance. For the original data, the mean distance to the centroid was 13.3 ± 0.36 , with a variance of 115.6 ± 5.5 , indicating widespread variability. After removing outliers, the mean distance decreased to 12.35 ± 0.2 , and the variance dropped to 33.3 ± 1.6 . This reduction in variance points to a more concentrated cluster around the centroid, suggesting that the outliers contributed to the wider spread in the original data, see fig. 4.14. The new, tighter cluster reflects a more defined grouping of points, now closer to the central tendency. After filtering for outliers, the ^{10}Be cluster still shows a residual trace on the left side. This is believed to originate from background noise rather than genuine ^{10}Be ions. However, it remains included in the total count due to limitations of the clustering algorithm.

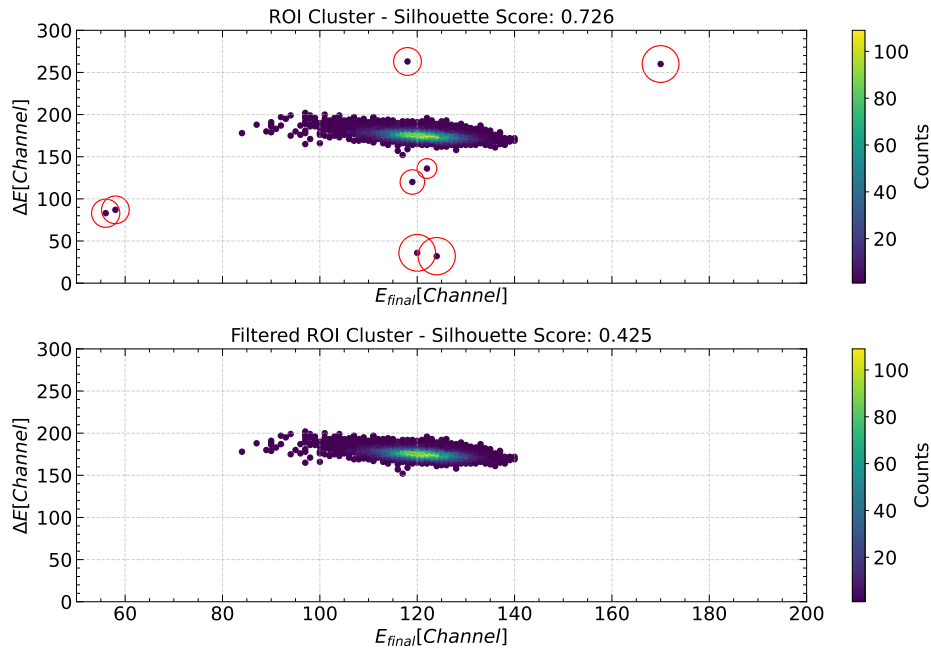


Figure 4.14: Comparison of ROI before(top) and after(bottom) filtering

The confidence value decreased slightly after filtering for outliers but still offered meaningful insight into the stability of the cluster structure post-removal, confirming that the filtering process enhanced cluster cohesion.

Furthermore, the total average number of counted points in the unfiltered cluster was 1892.6 ± 43.5 , while the filtered ROI cluster for ^{10}Be , shown in fig. 4.14, contained 1819.4 ± 43.5 . This indicates that the removal of outliers did not significantly reduce the dataset size compared to previous measurements. These points, identified by their indices, exhibited atypical behaviour or showed characteristics more consistent with other clusters, and were therefore excluded as irrelevant to the final analysis.

The 69% reduction in variance and the tighter clustering observed in the filtered data underscore the effectiveness of the OKOH algorithm in enhancing the precision and reliability of ^{10}Be detection.

Due to experimental constraints and limited measurement time, the algorithm was tested on only four samples, each consisting of ten runs.

The results demonstrate that the OKOH algorithm and the standard (normal) counting method yield consistent average isotopic ratio efficiencies of approximately 16.07 ± 0.10 and 15.94 ± 0.22 , respectively, with overlapping error margins as shown in fig. 4.15 (left). This close agreement highlights the reliability and precision of both counting methods. The right panel shows the simulated distributions for both methods, revealing very similar shapes and spreads, which further supports their statistical consistency. These distributions were generated under the assumption of normality using a Monte Carlo simulation approach, where repeated sampling was used to model the expected variation in isotopic means based on sample size and estimated uncertainties.

These average values are derived from the measured $^{10}\text{Be}/^9\text{Be}$ ratios, where the total ^{10}Be counts, obtained by multiplying the average counts per run by the number of runs, are divided by the total number of ^9Be ions. The uncertainty in the ratio is primarily governed by the statistical variation in the ^{10}Be counts.

To assess the overall system performance, the measured ratios are compared to a nominal reference value, yielding the isotopic efficiency. In this work, the nominal $^{10}\text{Be}/^9\text{Be}$ ratio was taken as 2.71×10^{-13} with an uncertainty of 0.3×10^{-12} . The uncertainty in the efficiency reflects contributions from both the measured and nominal ratios.

Importantly, by working with ratios, many systematic uncertainties, such as detector response, beam current variations, and transmission losses, are largely cancelled out, as they affect both isotopes similarly. This enhances the robustness of the results and ensures that the observed agreement between methods reflects true consistency rather than instrumental biases.

Compared to traditional manual gating or standard statistical filtering, the OKOH method offers a more automated and objective way to isolate ^{10}Be . The significant 69% reduction in variance and improved cluster definition validate its effectiveness.

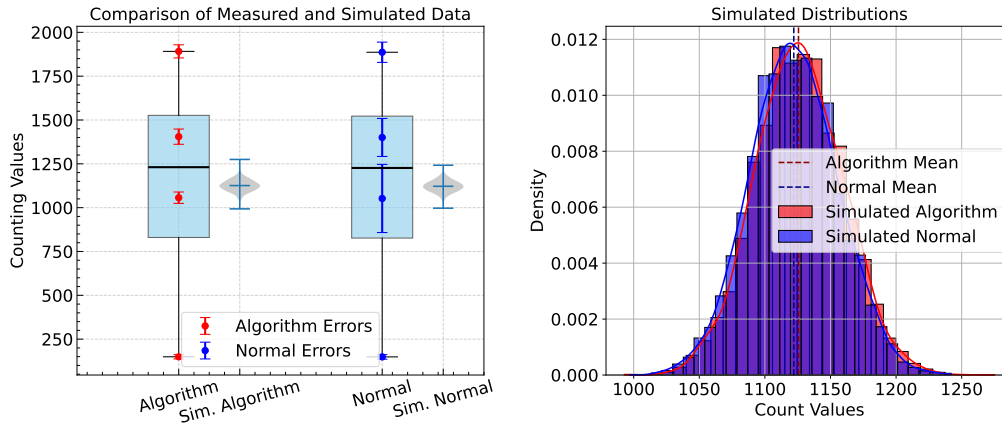


Figure 4.15: Comparison of average counted values from the Algorithm and Normal methods, obtained from appendix B (Degradation method). Both yield similar isotopic values with comparable precision. Left: Mean values with error bars; Right: Simulated distributions.

4.4 Si₃N₄ Foil Stack Methods Results

Using the Si₃N₄ foil stack method, we selected and analysed four spectra from a larger set corresponding to energies between 800 kV and 1000 kV. The gas pressure for these energies was set to approximately ~ 0.7 mbar to ensure detectable signal levels. This selection was necessitated by limited available data: Several required files were missing, and some spectra were unusable. Additional spectra were excluded due to poor quality, rendering them unsuitable for reliable analysis, see appendix B (Stacked Foil method). The files we were able to use are listed in table 4.1. These represent four usable spectra, out of the six usable ones, from a total of 34 available files.

Table 4.1: Summary of Si₃N₄ Foil Stack Data, all pressures where set at ~ 0.7 mbar.

File Name	Counts	Runs	Efficiency (%)
Be51_830kV	202	3	0.664 ± 0.028
Be51_840kV	2031	2	10.09 ± 0.194
Be51_850kV	20001	2	98.543 ± 1.197
Be51_880kV	232923	2	1299.08 ± 14.506

The stacked foil method using the Si₃N₄ foil stacks was intended to enhance detection efficiency, which under standard conditions typically reaches around 16%. However, the spectra obtained in this experiment exhibit numerous inconsistencies that severely limit any meaningful interpretation. The calculated efficiencies would span from less than 1% to over 1000%, which is non-physical and indicates underlying problems with the measurement conditions or data processing.

This may be attributed to the intentional reduction of both the tandem accelerator energy and the gas pressure in the GIC, prompted by concerns regarding the actual thickness of the Si₃N₄ foils.

The way these efficiencies were calculated was by using the same method as for section 4.3, but due to the behaviour of the data, we ended up counting

almost all of the points for each data file, making the algorithm useless for these measurements.

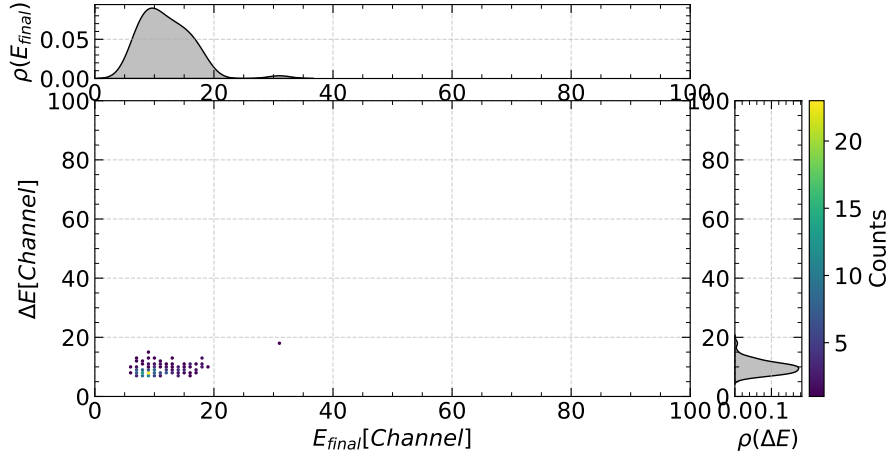


Figure 4.16: Spectrum from the Be51_830kV measurement.

As shown in fig. 4.16, the spectrum is quite small and low in total counts (202). It is visible that there is a peak, but it was not possible to determine whether or not these counts belong to ¹⁰Be ions at this energy. We believe that the structure is dominated by noise. This makes it unsuitable for extracting any useful information. This explains the low isotopic ratio efficiency, as very few detections were made in this run.

As seen in fig. 4.17, the distribution is slightly more populated than at 830 kV, with a total of 2031 counts. However, it still lacks a clearly defined peak, and the spectrum remains too diffuse to allow for confident analysis. There is noticeable skewness in the distribution, and it appears as though two overlapping components may be present, potentially producing two closely spaced shoulders rather than a single, dominant peak.

The centre plot in fig. 4.17 further supports this observation, showing a higher accumulation of counts in the lower-left region of the distribution, which may indicate a concentration of events at lower energy channels. Nonetheless, the lack of resolution and clarity prevents any definitive interpretation of these features.

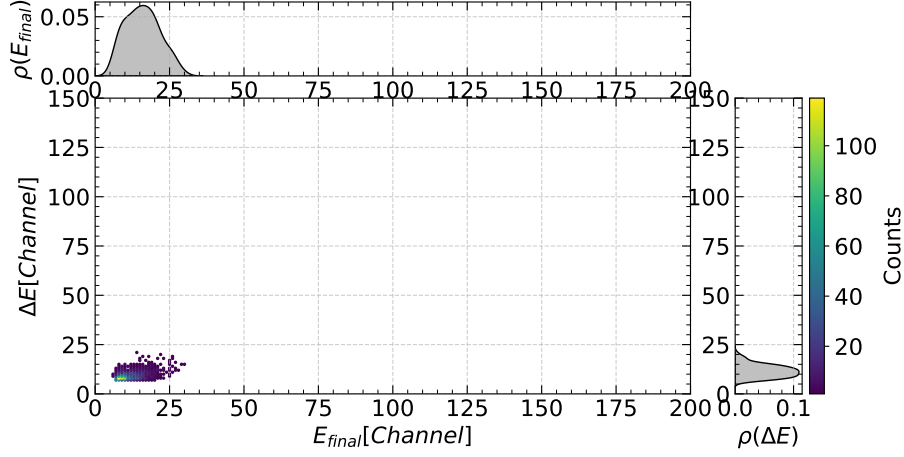


Figure 4.17: Spectrum from the Be51_840kV measurement.

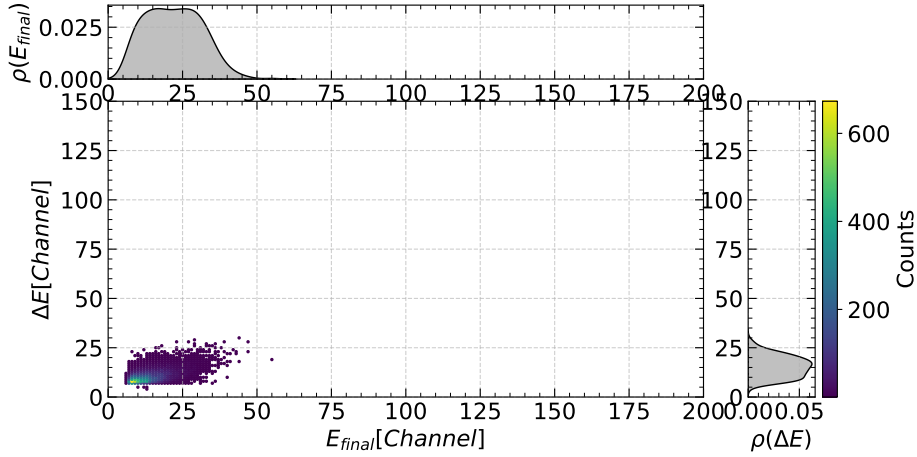


Figure 4.18: Spectrum from the Be51_850kV measurement.

At the energy level of 850 kV, shown in fig. 4.18, the spectrum contains a significantly higher count (20,001) and reveals a broader shape. There appear to be two broad maxima or shoulders in the distribution. However, these features are not separated or well-defined, making any further analysis uncertain and inconclusive.

The spectrum at 880 kV (fig. 4.19) shows an even more dramatic increase in total counts (232,923). Similar to the 850 kV case, the distribution suggests

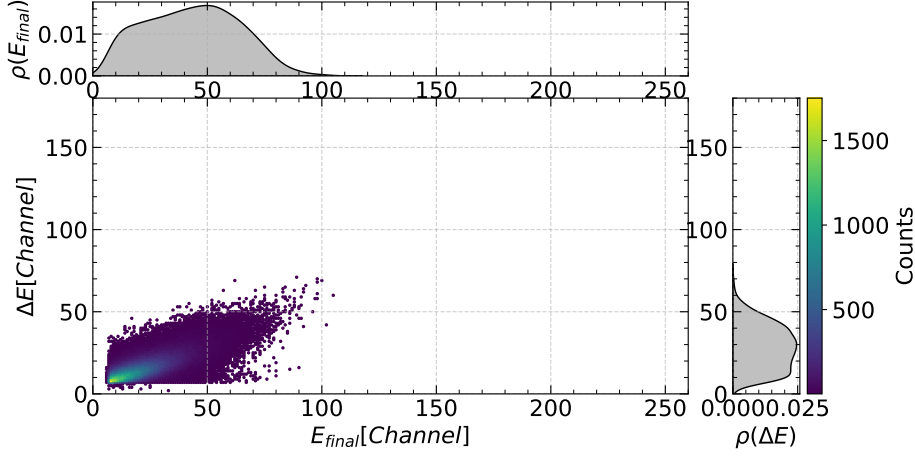


Figure 4.19: Spectrum from the Be51_880kV measurement.

the presence of two broad peaks, but the spectrum is overly compressed in the lower channel range and possibly saturated. These effects further obscure interpretation.

In the 850 kV to 880 kV spectra, the apparent presence of dual peaks is noteworthy. However, due to the poor resolution, elevated background, and general lack of clarity, these features cannot be reliably analysed or interpreted. The data does not support curve fitting or detailed modelling, so fitting for a distribution corresponding to ^{10}B and ^{10}Be ions at these energies were not possible.

Although the stacked foil method failed to separate ^{10}Be and ^{10}B at 1 MV, the TRIM simulations (Section 4.1) and analytical model (Section 4.2) indicate that the approach could within the correct conditions be applicable, and succeed at higher energies, where the difference in ion ranges becomes more pronounced. This aligns with the success of the method at 3 MV reported by Steier et al. (2019).

Chapter 5

Summary and Conclusion

This section summarises the key findings of the study and discusses their implications for AMS-based ^{10}Be detection.

5.1 Summary

To summarise the results and discussion, this thesis compared two methods for the detection and separation of ^{10}Be in AMS, the established Degradation method and an alternative approach using Si_3N_4 foil stacks. Each method was evaluated in terms of data quality, isotopic separation, and calculated detection efficiency.

The Degradation method produced clear and interpretable results. The spectra showed well-separated distributions for ^{10}B and ^{10}Be isotopes, allowing confident identification of ^{10}Be . This visual separation made it possible to interpret the results confidently. The algorithm developed for data analysis performed reliably and yielded an average isotopic detection efficiency of $16.07 \pm 0.1\%$, consistent with both internal benchmarks at AARAMS and values published by other AMS facilities such as VERA.

In contrast, the stacked foil method yielded inconclusive results. Only a small subset of the collected data could be analysed due to missing files and poor-quality spectra. The usable spectra exhibited a wide range of calculated efficiencies from below 1% to over 1000%, which are physically

unrealistic. These inconsistencies suggest substantial issues with either the measurement conditions or the experimental setup, which we were unable to investigate due to time constraints fully. The limited measurement time was further reduced by complications with the accelerator instruments and delays caused by the foil holder being delivered with incorrect dimensions.

Visual inspection of these spectra revealed irregular and often ambiguous structures. Some showed broad or overlapping peaks, while others were dominated by noise or saturation effects. These characteristics rendered standard analysis methods ineffective. Even where structure was present, as in the formation of different tops, the resolution was too poor to distinguish between ^{10}Be and ^{10}B as well as background signals confidently.

5.2 Conclusion

In conclusion, the algorithm developed for this study performed as intended. It enabled the reliable identification of isotopic signals in the degrader method and allowed for simulations that produced meaningful efficiency values. The consistency of these results with previous measurements demonstrates the robustness of both the technique and the analysis procedure. However, the stacked foil method using Si_3N_4 foil stacks did not yield usable results under the conditions tested. Due to experimental limitations, poor spectral quality, and instrumental issues, the data collected with this method were inconclusive under the tested conditions. Further optimisation and systematic testing would be necessary before this approach could be considered a viable alternative for ^{10}Be detection for the 1 MV at AARAMS. Additionally, due to time constraints, the proposed analytical method was not tested during this study. Its evaluation remains a potential direction for future investigation.

5.3 Outlook and Future Work

Extended Statistical Analysis: The OKOH algorithm demonstrated promising results for identifying low-count ^{10}Be signals in noisy datasets, but further refinement is needed. Future work should apply this method to larger datasets acquired under optimised experimental conditions, particularly focusing on improving its sensitivity to true ^{10}Be events while suppressing artefacts. Eventually, an alternative method for the detection of clusters can be used, the expectation–maximisation algorithm.

Experimental Validation of Analytical Model: Critical next steps include experimental verification of the analytical model’s predicted optimal parameters: A Si_3N_4 foil thickness of $1.5\,\mu\text{m}$ and a gas pressure of $12.2 \pm 5.45\,\mu\text{m}$ while running the accelerator at 1 MV. Controlled tests should specifically examine whether these parameters achieve the theorised improvement in ^{10}B suppression while maintaining adequate ^{10}Be transmission, as suggested by the energy-loss constraints. Such validation would confirm the model’s practical utility for low-energy AMS systems.

Foil and Instrument Optimisation: The current foil holder design requires tests to address potential alignment issues noted in which may contribute to beam scattering and data inconsistencies. Additionally, developing a diagnostic method to monitor foil effects in situ would help identify and correct any holder-related anomalies during experiments. A new pressure gauge would be practical as it would remove any concerns regarding the actual pressure.

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Appendix A

Beryllium Isotope Data

Table A.1: Standard $^{10}\text{Be}/^9\text{Be}$ Ratios for samples used.

Sample	$^{10}\text{Be}/^9\text{Be}$ (min)	$^{10}\text{Be}/^9\text{Be}$ (max)
Be54	4.77E-13	5.01E-13
Be51	4.53E-13	4.99E-12
Be62	4.75E-12	9.69E-14
Blank	$\sim 1\text{E-}16$	

Appendix B

Additional Figures

Degrader Method

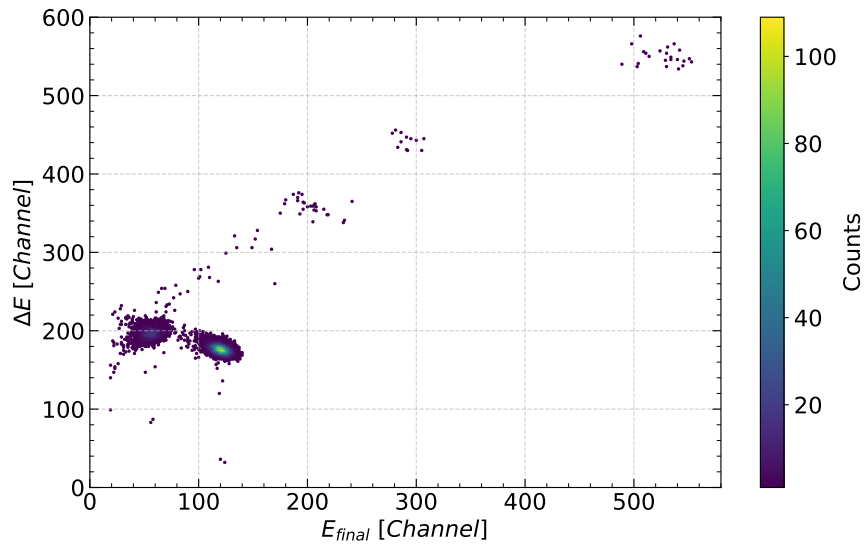


Figure B.1: Be51 sample spectrum using degrader method at 1 MV terminal acceleration voltage

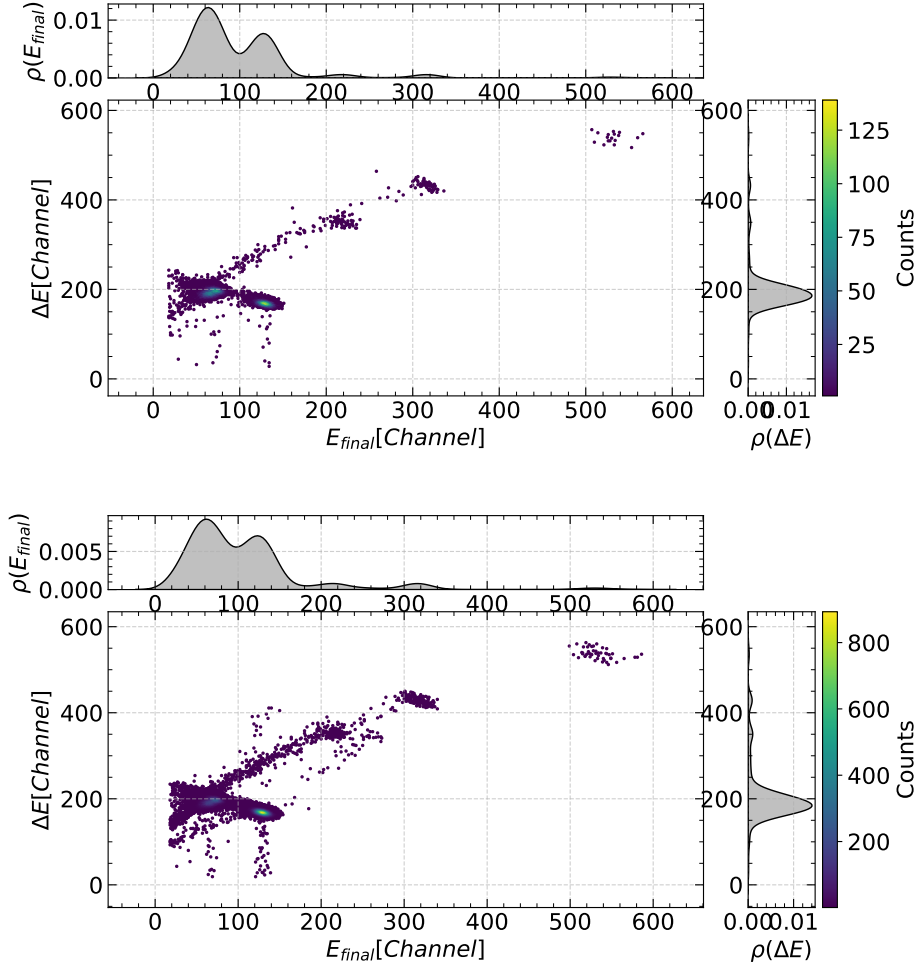


Figure B.2: Be54 sample spectrum using the degrader method at 1 MV terminal acceleration voltage, representing data from 120 runs.

Stacked Foil Method

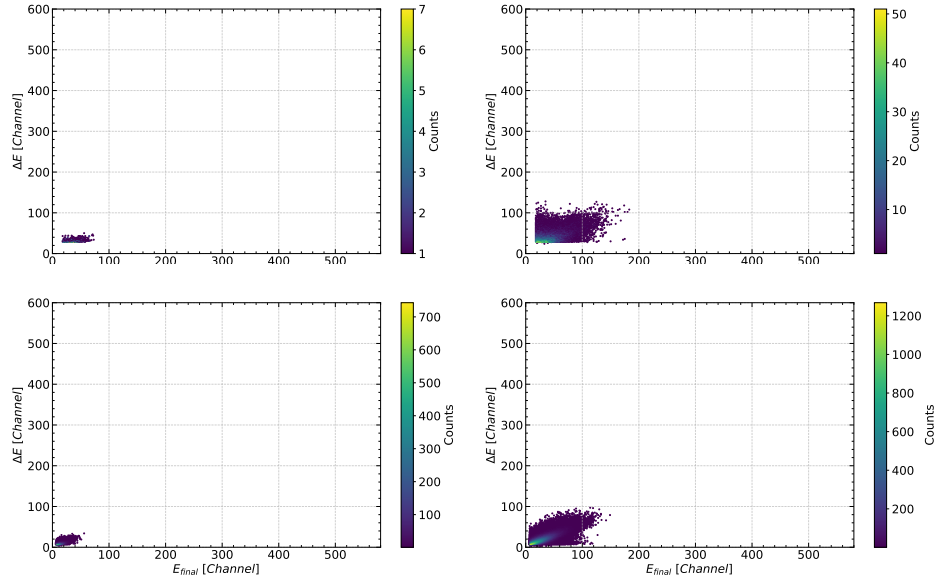


Figure B.3: Different spectra from Be51 (Top left, Bottom left, Bottom right) and Be54 (Top Right) obtained with the stacked foil method. Showing the complexity of getting a usable spectrum. These are at different energies